Adaptive Two-Stage Multisensor Convolutional Autoencoder Model for Lossy Compression of Hyperspectral Data

Jannick Kuester, Wolfgang Gross, Simon Schreiner, Wolfgang Middelmann, and Michael Heizmann

***Abstract*— The growing availability of hyperspectral remote sensing data, specifically from the new hyperspectral satellite missions, requires efficient data compression due to limitations in bandwidth and available storage space while simultaneously preserving the spectral characteristics. Machine learning approaches are a powerful way to address this challenge, but they are usually tailored to only work on one specific sensor. This work addresses the challenge of transferability of autoencoder (AE) models for lossy compression of spatially independent and unknown hyperspectral datasets acquired from different sensor platforms. We propose the CompNext1D, an advanced multistage adaptive network based on the architecture of the adaptive 1-D convolutional AE (A1D-CAE). The characteristic of the A1D-CAE allows pretraining on a large dataset with wide spectral variability and transferability to other sensor data, e.g., with potentially limited data availability. The compression performance of the CompNext1D is enhanced by using image statistics and shows a high degree of transferability to unknown spectral signatures. We evaluate the reconstruction accuracy with three experiments of increasing complexity. The evaluation is based on the reconstruction accuracy using the spectral angle (SA), signal-to-noise ratio (SNR), peak SNR (PSNR), and structural similarity index measure (SSIM) metrics, and the results are compared to other learning-based lossy compression techniques. We demonstrate the high transferability and generalizability of our A1D-CAE and CompNext1D for compression rates from $c_R = 4$ to $c_R \approx 100$ on hyperspectral data from different sensors and carrier platforms. The CompNext1D architecture performs well in compressing hyperspectral data from multiple sensor sources with different characteristics while achieving higher reconstruction accuracy compared to state-of-the-art methods.**

***Index Terms*— Autoencoder (AE), dimensionality reduction, entropy coding, feature extraction, hyperspectral imaging (HSI) compression, spectral analysis, spectral compression.**

I. Introduction

HYPERSPECTRAL sensors measure the reflected electromagnetic spectrum of an area in hundreds of narrow and contiguous wavelength intervals called bands. The captured HSI is available as a 3-D data cube consisting of two spatial dimensions and one spectral dimension. Many remote sensing-related applications, such as land cover classification [1], [2], [3], vegetation monitoring and analysis [4], [5], soil and mineral mapping [6], [7], military and defense applications [8], [9], and oil detection [10], [11], have a high demand for hyperspectral data. These applications benefit from the additional spectral information of hyperspectral data and their material-specific features. In Earth observation, hyperspectral data are typically measured by sensors mounted on drones, airplanes, or satellites. While airborne sensors cover only a limited area of the Earth's surface, satellite-based sensors collect information over the entire globe. Since a single hyperspectral dataset can be hundreds of megabytes in size, all three platforms require efficient data transmission and storage. This data volume complicates the downlink to a ground station, as bandwidth is limited and cannot be expanded without significant cost, especially for drone- and satellite-based systems. Transmitting large amounts of data is also essential for real-time applications, such as emergency management [12] and military applications. Hyperspectral data compression can be used to solve this problem, distinguishing between lossless and lossy compression. The compression ratio, denoted as c_R, achieved by lossless algorithms is limited to a factor of three [13]. This is insufficient for real-time hyperspectral data transmission with existing systems. Therefore, the focus is on lossy compression methods, with the challenge of finding a tradeoff between compression rate and reconstruction accuracy.

In recent years, many algorithms have been proposed for hyperspectral imaging (HSI) compression. Due to its high compression rate compared to lossless methods, lossy image compression has become an increasingly favorable choice. Lossy compression methods can be divided into signal processing-based methods and machine-learning methods. Most state-of-the-art signal processing methods for lossy compression are based on transform coding (TC), which transforms hyperspectral data into a less correlated, low-dimensional space. The transformed representation is then quantized, resulting in a loss of information [14], [15], [16]. Methods such as the 3-D discrete wavelet transform (DWT) [17] and the 3-D discrete cosine transform (DCT) [18] exploit the 3-D properties of HSI data. Wavelet-based transformations and significance map coding techniques further enhance the use of HSIs. These techniques, originally developed for 2-D images, have been adapted for 3-D coding. The embedded

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zerotree wavelet (EZW) [19], set partitioning in hierarchical trees (SPIHT) [20], and set partitioning embedded block coder (SPECK) [21] algorithms improve DWT performance by extracting the significance map.

In hyperspectral data, correlations between bands typically exist over the entire wavelength range, while spatial correlations exist only for a few pixels. 1-D transforms, such as the Karhunen–Loève transform (KLT) and principal component analysis (PCA), are used for spectral decorrelation in combination with a 2-D transform, namely, JPEG 2000, for spatial decorrelation, and an entropy coder. The combination of these 1-D and 2-D methods has proven to be an effective solution [22], [23], [24], [25]. Numerous publications have shown that PCA-based methods outperform DWT in terms of reconstruction accuracy and compression performance for the spectral dimension [26], [27]. The PCA transformation projects the input data onto a latent space spanned by a reduced number of uncorrelated PCs with the highest variance. Less significant, noisy PCs with lower variance are truncated to achieve 1-D spectral compression using only the statistically relevant PCs. Early practical use of PCA was limited by its high computational complexity, but the development of low-complexity PCA has mitigated this problem [28], [29]. However, the limitation of PCA's linear transformation in dealing with nonlinear correlations remains an issue. Therefore, lossy compression algorithms for hyperspectral data should consider both linear and nonlinear correlations to preserve as much information as possible in fewer components.

Machine learning-based compression methods, especially those using autoencoders (AEs), outperform classical signal processing techniques, including JPEG and JPEG 2000, in image compression performance [30], [31], [32]. Due to the compelling performance of AE approaches in image compression, they are increasingly used in hyperspectral data processing. One of the first applications of AE in the hyperspectral domain was dimensionality reduction and feature extraction [33], [34], [35]. It serves as a preprocessing step and counteracts the curse of dimensionality, e.g., to improve classification results [36]. Convolutional neural networks (CNNs) are also used for dimensionality reduction, which is further compressed using classical DCT and entropy coding [37]. A nonlinear PCA (NLPCA)-based multilayer perceptron with an AE architecture and nonlinear activation functions in the hidden layers is used to compress the spectral dimension of hyperspectral data [38], [39]. Due to the nonlinear properties of NLPCA and its fixed number of components, the information of the original spectrum is preserved and equally distributed among all components. A deep-layer, fully connected AE approach is used to compress the spectral dimension by representing the information in fewer learned significant features, with an emphasis on minimizing reconstruction error [40]. Reconstruction accuracy has been further improved by using a 1-D convolutional AE (1D-CAE) with higher feature learning capacity based on the wavelength invariant property [41]. However, these lossy compression methods prioritize high spectral reconstruction accuracy, leaving the spatial dimension unaddressed.

Several lossy compression approaches incorporate the spatial dimension into the AE architecture to exploit additional information [42]. However, these methods often smooth structural features in both the spectral and spatial dimensions due to information loss. In most cases, low-frequency information is preserved and does not capture enough structural information, especially at low bit rates, resulting in poorer reconstructions. The machine learning-based architecture, called CENet, includes a compression neural network, an edge extraction neural network, and an interactive dual-attention module to improve the structural representation of latent features [43]. The interactive dual-attention module helps the edge extraction network focus on moderate boundaries, resulting in more efficient use of bitrates and improved reconstruction quality. The spectral signals compressor network (SSCNet) is based on a deep convolutional AE that focuses on compressing the spatial dimension of hyperspectral data [44]. Satellite-based hyperspectral data are used to evaluate the SSCNet, and reconstruction accuracy is measured using the peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) metrics. Another lossy HSI compression algorithm, which we refer to as SpatialNet, is based on a 2-D convolution layer and a max-pooling layer to reduce the spatial dimension, followed by entropy coding using an adaptive arithmetic coder for transmission or storage applications [45]. While 3D-CNNs have been used in various HSI applications, they have not been extensively used in hyperspectral data compression. A 3D-CAE for an end-to-end optimization for both compression and reconstruction was introduced in [46]. The 3D-CAE slides across the hyperspectral data's spatial and spectral dimensions, providing an accurate joint spectral–spatial compressed representation. In addition, the loss function consists of weighted proportions of the PSNR, SSIM, and spectral angle (SA) metrics.

So far, the previously introduced compression methods have been applied and evaluated on a very limited amount of hyperspectral data, which is highly localized and, thus, spatially contiguous. Under these conditions, the transferability of the models cannot be fully assessed due to the limited spectral variability between training and test data. Since the most commonly used hyperspectral datasets are small and not well suited for training modern machine learning methods, satellite-based hyperspectral missions are a suitable data source. The hyperspectral space missions, such as PRecursore IperSpettrale della Missione Applicativa (PRISMA) [47], DESIS [48], and EnMAP [49], are free for scientific use. Access to the PRISMA hyperspectral satellite data provides a large amount of data with very high spectral variability for training machine learning-based methods. These data can be used directly without a complex labeling process since the main task of data compression is to reconstruct the original spectral signatures, and no labels are required. Furthermore, the variability of the data allows for spatial and temporal independence and separation of the evaluation datasets.

Another problem that has arisen with the established lossy methods based on machine learning is the transferability of the models to other hyperspectral data recorded by different

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sensors with varying characteristics. To solve this problem, the static model needs to be extended to allow compression of data from multiple sources. The transferability, combined with access to large amounts of data, enables the construction of backbone models with high spectral variability. This allows the models to be pretrained on a wide variety of data and fine-tuned on a target sensor-specific dataset with a deviating number of spectral bands.

In this article, we propose CompNext1D as an advanced multistage lossy hyperspectral compression model that is transferable across multiple sensors and still fine-tunable to specific sensor capabilities. The CompNext1D architecture consists of the adaptive 1D-CAE (A1D-CAE), the k-means quantization, and Huffman coding ensembles shown in Fig. 1. Due to its multistage characteristic, higher compression rates are possible, while the computation time stays fast enough for real-time applications. Caused by the solid understanding of large datasets and the spectral-based training, the compression results in a higher reconstruction accuracy. The main contributions of this article, motivated by the reduction of high-volume hyperspectral data and the increase in the efficiency of data transmission, storage, and onboard analysis, are given as follows.

1) *Transferability Across Sensors:* The adaptive extension of the A1D-CAE [41] model allows the compression of spectral signatures from different hyperspectral sensors with deviating numbers of bands and characteristics. This transferability eliminates the requirement for individual models for each sensor.
2) *Efficient Use of Large Datasets:* The adaptive characteristic enables pretraining on a large dataset with wide spectral variability. Through this, the model gets a solid understanding of hyperspectral data and learns more representative features, which enhances its robustness for reconstruction and generalization ability.
3) *Fine-Tuning Capabilities:* The pretrained model can be fine-tuned to fit datasets with different characteristics. This fine-tuning process ensures that the model's reconstruction accuracy is optimized for the unique characteristics of the target data.
4) *Multistage Compression:* The integration of a two-stage compression process, encompassing k-means quantization followed by Huffman coding, utilizes the image statistics of the compressed data. This two-stage compression increases the compression rate with minimal loss of reconstruction accuracy.
5) *Spectral-Based Compression:* With the same database, the CompNext1D model has more training samples available for the training process compared to 2-D- or 3-D-based models. This increased availability of training data ensures a more stable training process, resulting in a model with broad spectral variability and higher reconstruction accuracy.
6) *Real-Time Compression:* The model has a pioneering ability to compress hyperspectral data during capturing. This allows simultaneous data transmission, immediate onboard compressed analysis, and faster and more frequent data access.
7) *Performance Evaluation:* The proposed CompNext1D and the base model A1D-CAE are compared to state-of-the-art lossy hyperspectral compression methods. The reconstruction error is evaluated using several metrics, namely, SA, the signal-to-noise ratio (SNR), PSNR, and SSIM, for six unknown spatially and temporally independent datasets.

The remainder of this article is organized as follows. Section II introduces the theoretical background and the related methods required for the proposed approach. Section III introduces the experiments, the comparison metrics, and the reference methods. Section IV shows the results of the compression performance of the experiments. Finally, Section V presents the conclusions.

II. Proposed Method

This section introduces the basic architecture of the proposed A1D-CAE and describes the implementation of sensor transferability. The architecture of the A1D-CAE model is extended to CompNext1D by an additional compression stage. The additional compression stage increases the compression rate based on the image statistics. The compressed data of the first stage A1D-CAE are then further compressed in the second stage by adaptive entropy coding, as shown in Fig. 1. The dashed line in Fig. 1 also shows that A1D-CAE can be used as an independent compression method with a reduced compression rate.

A. Brief Review of AE

AEs are neural networks with the ability to learn efficient representations of input data, allowing unsupervised learning of complex data distributions. They are particularly useful in scenarios where labeled data are not available. In these cases, AE learns representations of the data without requiring explicit labels, enabling them to discover patterns and structures in the data that may be difficult to identify using traditional supervised learning approaches. The AE architecture can be divided into an encoder and a decoder. Since the spatial dimension is not used in our approach, the 3-D hyperspectral dataset can be reorganized as a 2-D matrix with the following transformation $\mathbf{H} \in \mathbb{R}^{m\times n\times b} \rightarrow \mathbf{V} \in \mathbb{R}^{p\times b}$, with $p = m \cdot n$ pixels and b bands. The input of the encoder is a single spectral signature $\mathbf{x} \in \mathbb{R}^{b}$. The encoding tasks of finding and extracting the essential features are individual transformations of each layer, concatenated into an overall transformation for the encoder $f : \mathbb{R}^{b} \rightarrow \mathbb{R}^{\hat{b}}$. The encoder transforms the spectral signature into a low-dimensional representation

$$\hat{\mathbf{x}} = f(\mathbf{W_E}, \mathbf{x}) \in \mathbb{R}^{\hat{b}} \tag{1}$$

where $\mathbf{W_E}$ are the encoder weights and $\hat{b} \ll b$ is the dimension of the compressed representation. Lossy compression is realized by the transformation $f : \mathbb{R}^{b} \rightarrow \mathbb{R}^{\hat{b}}$, which uses fewer features to represent the input spectral information. The decoder reconstructs the compressed signal with the corresponding transformation $g : \mathbb{R}^{\hat{b}} \rightarrow \mathbb{R}^{b}$. The decoder transforms the compressed spectral information back to the original representation according to

$$\tilde{\mathbf{x}} = g(\mathbf{W_D}, \hat{\mathbf{x}}) \in \mathbb{R}^{b} \tag{2}$$

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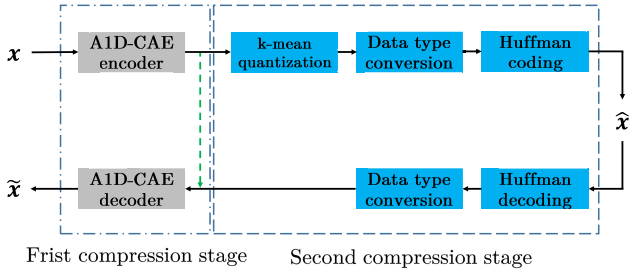


Fig. 1. Overview of the two-stage architecture of the CompNext1D. The first stage contains the A1D-CAE and can also be used independently. The second stage, consisting of k-means quantization, data type conversion, and Huffman coding, increases the compression rate of the A1D-CAE based on the image statistics.

where $\mathbf{W_D}$ are the decoder weights. The goal of the inverse transformation is to reconstruct the spectral signature as accurately as possible by minimizing the error between the input data and its reconstructed version.

B. A1D-CAE

The increasing availability of large-area unlabeled hyperspectral data from space missions and the fact that labeled data are not required favor the use of large and deep networks. To take advantage of the large amount of data available, we designed the A1D-CAE, as shown in Fig. 2, which is based on an AE structure. The A1D-CAE compresses the spectral dimension with a focus on high spectral reconstruction accuracy. Thus, the input to the model is a single spectral signature $\mathbf{x} \in \mathbb{R}^b$. In order to use the large amount of data available from space missions for other hyperspectral sensors, the model needs the property of sensor transferability. We have implemented adaptive padding so that the A1D-CAE can process spectral signatures from different sensors with different numbers of bands. The main objective of padding is to pad the spectrum, so it is divisible by the specified compression ratio. An important aspect that must be considered when filling is the physical property, whereby the recorded values of the spectral signature exceed 0. Reflection padding is employed to prevent discontinuities created by the padding. This strategy can help minimize discontinuities at the spectrum's edge and ensure that populated values correspond to true values within the spectrum, stabilizing the training process. The model additionally needs the desired compression rate c_R and the number of bands of the spectral signature as parameters to calculate the required number of padding bands as

$$\Delta_{\text{pad}} = \left(\left\lceil \frac{b}{c_R} \right\rceil \cdot c_R \right) - b. \tag{3}$$

The resulting spectral signature of $b + \Delta_{\text{pad}}$ is now divisible by the compression ratio c_R without residual error. The padding step depends on Δ_{pad} and is needed for $\Delta_{\text{pad}} > 1$ in order to reconstruct the spectral signature with the original spectral dimension $\tilde{\mathbf{x}} \in \mathbb{R}^b$. Within the 1AD-CAE model, the b values for all bands are referred to as features because the physical properties of the bands are lost in the transformations. The basic architecture of the encoder and the decoder consists of six modules without the padding module, as shown in Fig. 2. Each module is composed of blocks with different

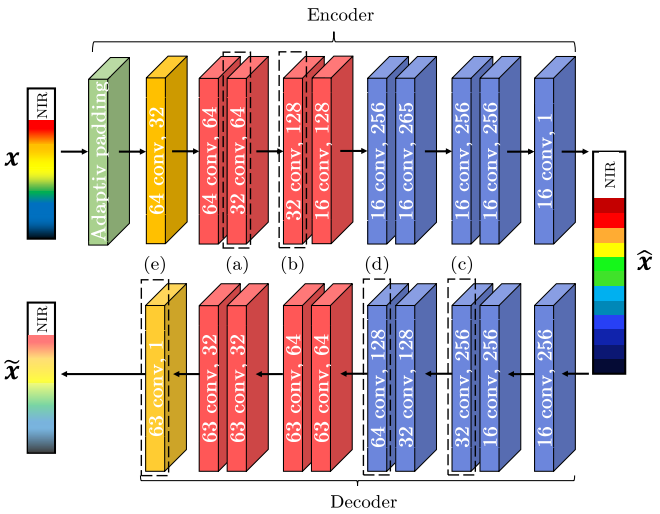


Fig. 2. Overview of the architecture of the A1D-CAE for the compression rate $c_R = 4$ and a spectral signature with $b = 63$ bands. The encoder has a total of seven modules, including the padding module. The first number indicates the number of bands, and the last number indicates the number of channels. The different colors visualize the number of kernels; yellow corresponds to a kernel size = 11, red = 9, and blue = 5. The functionality of blocks (a)–(e) is described in more detail in Fig. 3.

tasks and characteristics. In Fig. 2, module (a) uses blocks that are exclusively responsible for finding features, as shown in Fig. 3(a). Modules that have a block with downsampling properties, such as module (b) detailed in Fig. 3, reduce the number of features by a factor of two. The number of downsampling blocks (DBs) is determined by

$$\Delta_{\text{DB}} = \log_2(c_R). \tag{4}$$

Upsampling blocks (UBs) are only used in the decoder in the different modules, as shown by the dashed box (c) in Fig. 2. The UB, as shown in Fig. 3(c), doubles the number of features from the previous block. The reconstructed features are learnable. The architecture of the encoder is reflected in the decoder except for the adaptive padding layer. From the mirroring, it follows that #UB = #DB. Each UB in the decoder part reconstructs twice as many features as the previous block. Through adaptive padding, compression rate-dependent block configuration, and the use of 1-D convolutional layers, A1D-CAE is able to process spectral signatures from different hyperspectral sensors.

Next, we introduce the idea behind the design of the A1D-CAE modules, as shown in Fig. 2. Each module consists of at least one block composed of three layers. The block starts with a 1-D convolution layer or a 1-D transposed convolution layer, as shown in Fig. 3, which has the following task, depending on the parameters:

1) feature finding [see Fig. 3(a)];
2) feature extraction [see Fig. 3(b)];
3) feature reconstruction [see Fig. 3(c)];
4) feature reconstruction with cropping [see Fig. 3(d)];
5) scale reconstruction to [0, 1] [see Fig. 3(e)].

The 1-D convolution layer is followed by the batch normalization layer and ends with the GELU activation function, except for the last block in the decoder [see Fig. 2(e)]. Multiple channels are used in parallel to find and extract useful

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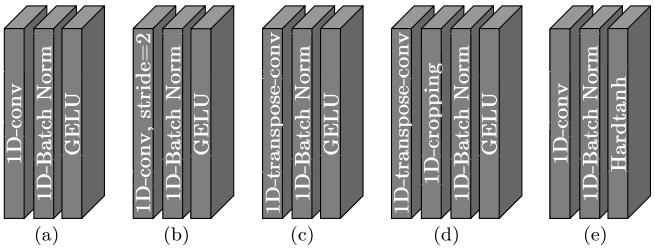


Fig. 3. Different blocks used in the architecture of the A1D-CAE. Block (a) is responsible for finding features, block (b) for feature extraction, block (c) reconstructs features, block (d) has an additional cropping layer, and block (e) is used to scale the value range of the reconstruction to [0, 1].

features for the compression process. The channel parameter in convolutional layers determines the number of output feature maps, allowing the network to recognize different patterns in the input data. The number of channels is represented by the second number in each module in Fig. 2. The number of channels is gradually increased to 256 in order to capture complex relationships between patterns, even at increasing depth, and is a tradeoff between computational complexity and reconstruction accuracy. The kernel size is an essential hyperparameter for the A1D-CAE, influencing both feature extraction and reconstruction. It determines how many bands the A1D-CAE considers during feature calculation. The kernel size parameter is chosen depending on the number of features in the module. The first module has a larger kernel size to detect and exploit long-range spectral correlations and broad patterns of features. As the number of features decreases in subsequent modules, the kernel size is reduced to focus on local patterns and reduce the required computational power. The number of kernels is represented by colors in Fig. 2. The downsampling operation in the DB is performed by the convolutional layer via the stride parameter shown in Fig. 3(b). This design choice is based on the fact that the extracted features are learnable compared to static downsampling performed by the pooling layer. In the final module of the encoder, the number of channels $\hat{c}_{\mathrm{BN}}$ is reduced to one, resulting in a compressed representation that comprises a limited set of essential learned features. The compressed data $\hat{\mathbf{x}} \in \mathbb{R}^{\hat{b}}$ can be derived from the output of the encoder, referred to as the bottleneck (BN). The compressed representation is determined by the number of features and channels present in the BN. The compression is the ratio of the size of the original spectral signature to the compressed representation and is calculated as

$$C_{\mathrm{R}} = \frac{b \cdot c_{\mathrm{in}} \cdot n_{\mathrm{Bit}}}{\hat{b} \cdot \hat{c}_{\mathrm{BN}} \cdot \hat{n}_{\mathrm{Bit}}} \tag{5}$$

where b and $\hat{b}$ denote the number of bands, c_{in} and $\hat{c}_{\mathrm{BN}}$ the number of channels, and n_{bit} and $\hat{n}_{\mathrm{bit}}$ the bit depths of the original and compressed signal.

The decoder mirrors the encoder structure, except that, in the UB, the 1-D convolutional layer is replaced by the 1-D transposed convolutional layer, as shown in Fig. 3(c). In addition, the last UB, marked by the dashed box, contains a cropping layer, as shown in Fig. 3(d). The last block of the decoder module uses the Hardtanh activation function, as shown in Fig. 3(e). The use of a 1-D transposed convolution layer makes the reconstruction process trainable and more adaptive. The cropping layer reduces the features by Δ_{pad}. The Hardtanh activation function is used to scale the range of the reconstructed spectral signature to [0, 1]. This is necessary to compare the reconstructed signal to the input signal and calculate the reconstruction error between the two spectral signatures.

Once the model is designed, the training process must be determined. The weight matrices $\mathbf{W_E}$ and $\mathbf{W_D}$ are initialized with random values. The loss function $\epsilon_{\mathrm{loss}}(\mathbf{x}, \tilde{\mathbf{x}})$ evaluates the reconstruction accuracy of the training process. We select the mean square error (mse) as ϵ_{loss} for our proposed method. After each epoch of training, the weight matrix of each layer is updated using the backpropagation algorithm in combination with the Adam optimizer [50]. The learning rate (LR) starts at 1×10^{-3} and is reduced by a factor of 0.9 after reaching a plateau, which occurs when the value of the loss function ϵ_{loss} does not significantly change for five consecutive epochs. We used a batch size of 1024 and trained the model for 250 epochs. The convergence of the loss function to a local minimum indicates a successful training process. Since this is not a convex optimization problem, convergence to a global minimum cannot be guaranteed. For the training process, only the A1D-CAE model of the CompNext1D method needs to be trained since the k-means quantization and Huffman coding are recalculated for each dataset to achieve higher reconstruction accuracy. After training the A1D-CAE model, the two-stage CompNext1D method is compiled. This means that A1D-CAE models with different compression rates can be directly combined with the second compression stage.

C. Adaptive Entropy Coding

In the second compression stage of the CompNext1D, the compression rate can be further increased by exploiting the statistics of the hyperspectral data. The right-hand side of Fig. 1 shows the optional part consisting of a vector quantization technique, namely, k-means quantization, followed by data type conversion and Huffman coding. The second stage processes a matrix $\hat{\mathbf{V}} \in \mathbb{R}^{\mathrm{p} \times \hat{\mathrm{b}}}$ composed of the A1D-CAE output $\hat{\mathbf{x}} \in \mathbb{R}^{\hat{\mathrm{b}}}$. The compressed spectral signatures from the A1D-CAE are of the datatype float32, which affects the compression rate according to (5).

First, the k-means quantization calculates adaptive binning overall values in $\hat{\mathbf{V}}$. The result provides the k values representing the data with minimal loss, and the values in $\hat{\mathbf{V}}$ are identified with their corresponding bin values. This step is performed to improve the performance of the Huffman coding. The resulting compression rate can be controlled by the number of clusters n_{cluster} selected for the k-means quantization. The lower the number of clusters, the higher the compression of the Huffman coding, resulting in a lower reconstruction accuracy.

The quantization step includes multiplying the compressed representation by 10 000 and then converting the data type to int16. This step may result in additional information loss. The converted data are then losslessly compressed using Huffman coding. The output of Huffman coding is the compressed data and a symbol codebook. After transmission, the data are decoded with the information from the symbol codebook using the Huffman coding technique. The result is converted to

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float32 and then divided by 10 000. The compressed representation is transferred to the A1D-CAE decoder. Due to the high abstraction capability and generalizability of the A1D-CAE, a higher compression rate is achieved with a comparatively small increase in reconstruction error.

III. Experiments

We evaluate the proposed method through a series of experiments. First, the A1D-CAE base model is trained on a set of PRISMA data and evaluated for data compression on different unknown data of the corresponding sensors. Second, the CompNext1D with the previous A1D-CAE model weights is used to investigate the influence of higher compressions achieved by using adaptive entropy coding on the same set of PRISMA test data. Third, we investigate the adaptive capabilities of A1D-CAE to unknown airborne and drone hyperspectral data from different sensors. The accuracy of the reconstruction is evaluated using the PSNR, SNR, SA, and SSIM metrics. We use public remote sensing HSI datasets from the PRISMA and DESIS missions, and the PaviaU benchmark dataset. In addition, we use datasets from our previous measurement campaigns with aisaEAGLE II and HySpex Mjolnir V-1240 hyperspectral sensors. We compare the proposed method with the state-of-the-art algorithms, PCA and NLPCA, and the two recently presented machine learning-based models, SSCNet and SpatialNet. In addition, the 3D-CAE is used as a reference method to analyze the validity of 3-D methods for hyperspectral data compression.

A. Datasets

In our experiments, we evaluate the performance of hyperspectral data compression on datasets collected from satellite, airborne, and drone platforms. These datasets have different characteristics in terms of spectral range, spatial/spectral resolution, processing level, and noise level.

1) *PRISMA:* The PRISMA Earth observation satellite of the Italian Space Agency ASI measures hyperspectral data around the world [47]. In May 2020, the satellite started scientific operations. The silicon-based detector system covers the visible and near-infrared range (VNIR) from 420 to 1010 nm with $b = 63$ bands. Data from the L2B processing level are used. The L2B level is radiometrically corrected and georeferenced with a ground sampling distance of 30 m. Each dataset contains 1000 $\times$ 1000 pixels. Floating-point 32-bit single precision is used as the data type.
2) *DESIS:* The DESIS hyperspectral sensor is part of the MUSES platform and installed on the International Space Station (ISS). The system has been operational since June 2018 [48]. The sensor covers the VNIR range from 400 to 1000 nm with $b = 235$ bands. Data from the L2A processing level are used. The datasets are corrected using a spectral reference of land-based objects and a ground sampling distance of 30 m. The image size per dataset is approximately 1500 $\times$ 1500 pixels. The data are provided in the 16-bit integer format.
3) *Greding:* The Greding dataset was acquired in 2014 in Greding, Germany. It was recorded with an aisaEAGLE II hyperspectral sensor with $b = 127$ bands, covering the electromagnetic spectrum from 390 to 990 nm. An airplane was used as a carrier platform flying at an altitude of 750 m. The data are radiometrically and atmospherically corrected and georeferenced with a ground sampling distance of 0.5 m. This 670 $\times$ 606 pixel dataset is freely accessible and has a manually labeled ground truth of six classes. The data are provided in the 16-bit integer format.
4) *Kleinhohenheim (KHH):* The KHH dataset was recorded in 2022 near Stuttgart, Germany. The measurements were performed with the HySpex Mjolnir V-1240, and a drone was used as a carrier platform. The hyperspectral sensor has $b = 200$ bands and covers the VNIR range from 410 to 990 nm. The datasets are radiometrically and atmospherically corrected and georeferenced with a ground sampling distance of 0.08 m. The georeferencing process rearranges the position of the flight strip in space and inserts additional zero spectra, where all bands have the value zero. The processed dataset has a spatial dimension of 3805 $\times$ 3940 pixels. In total, 1 591 610 million nonzero spectra are available for the evaluation experiments. After processing, HySpex stores the data in the 64-bit double precision (float64) floating-point format.
5) *University of Pavia:* The University of Pavia dataset was recorded in 2006 in Pavia, Italy. The dataset was acquired with the ROSIS-03 sensor covering the spectral range from 430 to 860 nm and consists of 340 $\times$ 610 pixels with $b = 103$ bands. The datasets are radiometrically corrected and georeferenced with a ground sampling distance of 1.3 m. The data are provided in the 16-bit integer format.

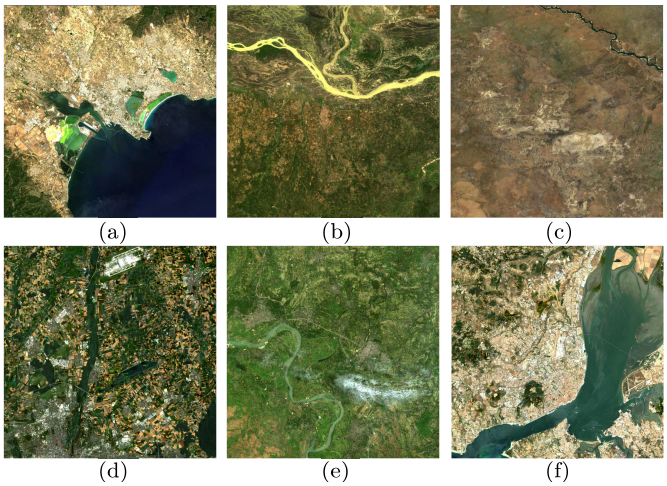


Fig. 4. Selection of the different training datasets from the PRISMA mission in RGB composition with bands $R = 32$, $G = 21$, and $B = 10$. The training datasets were collected over the following countries: (a) Italy, (b) Nigeria, (c) Sambia, (d) Germany, (e) India, and (f) Portugal, covering a wide range of spectral variability.

B. Evaluation Metrics

To evaluate the reconstruction accuracy, we use four common metrics, namely, the adjusted SNR, the PSNR, the SA, and the SSIM. Since lossy compression methods are used,

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it is important to measure the information loss based on the reconstruction error and from different perspectives.

SNR: It is measured between the input data and the reconstructed data. The SNR from [51] is adapted to hyperspectral data and calculated as

$$\epsilon_{\mathrm{SNR}}(\mathbf{V},\tilde{\mathbf{V}}) = 10\log_{10}\left(\frac{\sum_{k=1}^{p}\sum_{i=1}^{b}\mathbf{V}_{k,i}{}^{2}}{\epsilon_{\mathrm{mse}}\left(\mathbf{V},\tilde{\mathbf{V}}\right)}\right) \tag{6}$$

where the mse is defined as

$$\epsilon_{\mathrm{mse}}(\mathbf{x},\tilde{\mathbf{x}}) = d(\mathbf{x},\tilde{\mathbf{x}}) = \frac{1}{b}\sum_{i=1}^{b}(\mathbf{x}_{\mathrm{i}}-\tilde{\mathbf{x}}_{\mathrm{i}})^{2}. \tag{7}$$

The variable $\mathbf{V}$ is the original, and $\tilde{\mathbf{V}}$ is the reconstructed hyperspectral data. This SNR definition is useful for comparing different scenes with highly variable image statistics. The smaller the difference between the original and reconstructed datasets, the larger the SNR value and the higher the spectral similarity.

PSNR: It is measured between the input data and the reconstructed data. The PSNR from [52] is applied to the hyperspectral data and calculated as

$$\epsilon_{\mathrm{PSNR}}(\mathbf{V},\tilde{\mathbf{V}}) = 10\log_{10}\left(\frac{(\mathbf{V}_{\max}-\mathbf{V}_{\min})^{2}}{\epsilon_{\mathrm{mse}}\left(\mathbf{V},\tilde{\mathbf{V}}\right)}\right). \tag{8}$$

As with the SNR metric, the smaller the difference between the original and reconstructed datasets, the larger the PSNR value and the higher the spectral similarity.

SA: It is computed between the input and reconstructed spectral signatures to determine the reconstruction error [53]. The SA error is computed for each pixel individually and defined as

$$\boldsymbol{\epsilon}_{\mathrm{SA}}(\mathbf{V},\tilde{\mathbf{V}}) = \arccos\frac{\sum_{i=1}^{b}\mathbf{V}_{i}\tilde{\mathbf{V}}_{i}^{\intercal}}{\sqrt{\sum_{j=1}^{b}\mathbf{V}_{j}^{2}\sum_{l=1}^{b}(\tilde{\mathbf{V}}_{l}{}^{2})^{\intercal}}} \tag{9}$$

where $\mathbf{V}^2$ is defined as an elementwise square. $\mathbf{V}$ is the original, and $\tilde{\mathbf{V}}$ is the reconstructed spectral signature. The smaller the SA value, the higher the spectral similarity, which means less deviation between the original and the reconstructed signal. To compare the results, the mean SA value over all pixels is calculated.

SSIM: It is an image quality metric that evaluates the similarity between two images based on their structural information. It calculates the similarity of the luminance, contrast, and structure components between the two images by comparing the local neighborhoods of pixels. The SSIM ranges from -1 to 1, with values closer to 1 indicating greater similarity. The SSIM has been shown to be an effective image quality metric. The SSIM is used from the scikit-image Python framework [52] and applied on the original $\mathbf{V}$ and the reconstructed hyperspectral data $\tilde{\mathbf{V}}$.

C. Experiment Design

We investigate three different experiments. In the first experiment, we focus on generalizability and transferability to unknown hyperspectral data from different locations and at different acquisition times of the same sensor. The second experiment explores the effect of different compression rates and multistage compression. The third experiment examines the complex task of transferability of the A1D-CAE model to unknown data from different sensors with various characteristics.

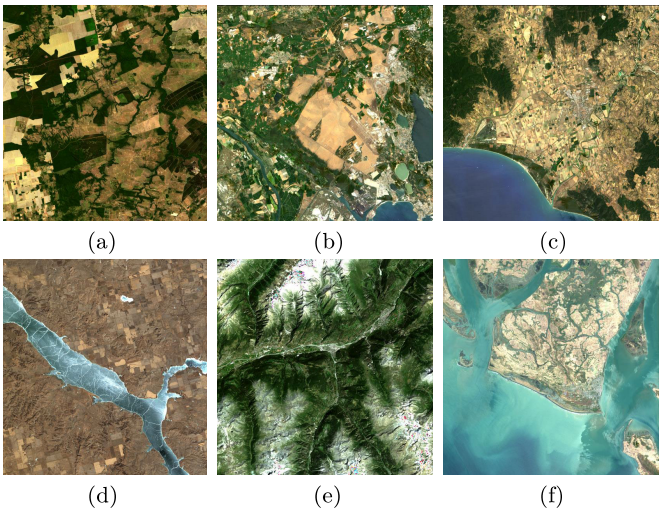


Fig. 5. Six test datasets from the PRISMA mission in RGB composition with bands $R = 32$, $G = 21$, and $B = 10$. The training datasets were collected over the following countries: (a) Brazil, (b) France, (c) Italy, (d) USA, (e) Switzerland, and (f) Myanmar, covering a wide range of spectral variability and, thus, a complex evaluation task.

1) Experiment 1: For the first experiment, we took 20 hyperspectral PRISMA datasets from around the world with a wide variety of landscapes. Their spectral signatures include different vegetation, agriculture, rural and urban areas, plains, deserts, mountains, and coastal regions with differently colored water bodies. Fig. 4 shows the RGB compositions of a selection of the training data. The exact names of the training datasets are listed in the Appendix. These datasets are used only for the training process and the validation step within the training. For the evaluation, we took six independent PRISMA datasets from different locations and different recording times than the training data. This ensures that the test data are completely unknown to the compression algorithms. Fig. 5 shows an RGB composition of the test datasets. The exact names of the test datasets are listed in the Appendix. This experiment uses the basic version of the A1D-CAE and is compared to state-of-the-art methods with compression rates of $c_R = 4$, $c_R = 8$, and $c_R = 16$ and without the second stage, including the adaptive entropy coding part.

2) Experiment 2: For the second experiment, we use the same training and test datasets as in the first experiment, as shown in Figs. 4 and 5. The base models are configured for compression rates of $c_R = 4$, $c_R = 8$, and $c_R = 16$. The compressed representations are further compressed with the second stage adaptive entropy coding part. The compression rate is experimentally adjusted by the number of clusters n_{cluster} from the k-means quantization algorithm to achieve compression rates of approximately $c_R \approx 32$, $c_R \approx 64$, and $c_R \approx 100$ for all methods. This allows A1D-CAE to be compared with PCA, NLCA,

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SSCNet, SpatialNet, and 3D-CAE. The second experiment is designed to evaluate the generalization ability and transferability to unknown test datasets from PRISMA for the extended compression method using the adaptive entropy coding part.

3) Experiment 3: The third experiment uses the same 20 training datasets from the PRISMA mission as the previous experiments. As test data, we use the DESIS, Greding, and KHH datasets and the PaviaU benchmark dataset. To recap, the test datasets are completely unknown to the compression method and have different characteristics, including the following:

1) number of bands;
2) ground sampling distance;
3) level of noise;
4) lighting influences;
5) atmospheric influences;
6) bidirectional reflectance distribution function (BRDF) effects;
7) preprocessing methods.

The test datasets from different sensors are illustrated as an RGB composition in Fig. 6. The third experiment cannot be evaluated using any of the reference methods. The state-of-the-art methods, PCA and NLPCA, are not applicable to the test datasets due to their characteristics and structure. Thus, they are not considered in this experiment. This is because the dimensions of the data are different from those of the training data and are, therefore, incompatible with the transformation matrix of the PCA. The fully connected layers of NLPCA prevent its application to the test data due to the different dimensions of the test data. The two lossy compression methods, SSCNet and SpatialNet, cannot compress datasets from different sensors without significant adjustments because the input layer always expects the same dimension as the training data. Therefore, only the A1D-CAE model with $c_R = 4$, $c_R = 8$, and $c_R = 16$ as compression rates without a second stage is evaluated.

D. Reference Methods

The reconstruction accuracy of the A1D-CAE and CompNext1D is compared to the other learning-based lossy compression methods, namely, PCA and NLPCA [38], SSCNet [44], SpatialNet with entropy coding [45], and the 3D-CAE [46]. The implementation of the PCA method is used from the Python framework sklearn [54]. All methods were trained on identical datasets with the same number of epochs. In addition, the SSCNet model uses tiling with a tile size of 64×64. The 3D-CAE is trained with the weighted loss function consisting of PSNR, SSIM, and SA, as described in [46]. The training process is stabilized by gradient clipping. Furthermore, a tiling size of $64 \times 64 \times 63$ is used. For the SpatialNet, we replaced the adaptive arithmetic coder part with the combination of the k-means quantization algorithm and the Huffman coding described in Section II-C in order to set the compression ratio more precisely and, thus, improve the comparability. The implementation of the k-means quantization algorithm is used from the Python framework scikit-learn [54]. All models are trained on the same NVIDIA Tesla V100 graphics processing unit (GPU).

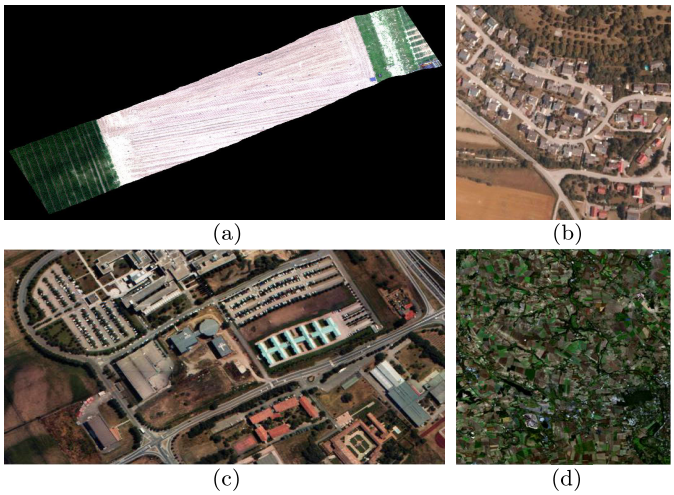


Fig. 6. Different test datasets as RGB composition from S III-C3. (a) KHH scene was acquired with a HySpex drone-based system ($R = 81$, $G = 50$, and $B = 19$). (b) Greding scene was acquired with the aisaEAGLE II airborne system ($R = 50$, $G = 30$, and $B = 15$). (c) PaviaU scene was acquired with an ROSIS-03 airborne system ($R = 50$, $G = 30$, and $B = 15$). (d) Scene of the DESIS test dataset was acquired with a satellite-based system ($R = 81$, $G = 50$, and $B = 19$).

E. Training Preparation

Prior to training, zero spectral values in the PRISMA data that result in vertical stripes in the data are replaced by inserting the average of their nonzero neighbors. Then, the value range of the hyperspectral dataset is scaled to $[0, 1]$, and the datatype is converted to float32. For the DESIS and the KHH datasets, the zero spectra inserted during the georeferencing process are removed, and the data are scaled to the value range $[0, 1]$. The Greding and PaviaU datasets are only scaled to the value range $[0, 1]$.

IV. Results and Discussion

A. Evaluation Experiment 1

We evaluate the transferability of the proposed A1D-CAE to unknown data of the same hyperspectral sensor and unknown data of other hyperspectral sensors based on the measured reconstruction error. Furthermore, we test the reconstruction accuracy of the CompNext1D for different compression rates. Table I shows the reconstruction error of the first experiment described in Section III-C1 for the compression rate $c_R = 4$, with the best results in bold. The proposed A1D-CAE outperforms the other compression methods for each metric and on all test datasets. The SA values of the A1D-CAE are far below 1° for all datasets. According to Licciardi and Chanussot [38], reconstruction errors of up to 3° indicate a high level of reconstruction accuracy. Since the SA values are less than 1°, we consider the loss of information to be small, indicating a very high reconstruction accuracy. Compared to the PCA and NLPCA, which also compress only the spectral dimension, the A1D-CAE has a 0.2°–0.5° lower SA value. For the reference methods, SSCNet and SpatialNet, both of which also take the spatial dimension into account during compression, the SA values are 2.6°–5.5° higher for SSCNet and even 7°–27° for SpatialNet under the same conditions. The 3D-CAE, which considers both spatial and spectral dimensional features, achieves SA values from 6.1°–13.7° and has a lower reconstruction accuracy than the A1D-CAE.

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TABLE I

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR EXPERIMENT 1 WITH COMPRESSION RATE $c_R = 4$ ON DIFFERENT PRISMA DATASETS BASED ON THE SA, SNR, PSNR, AND SSIM METRICS

$c_R = 4$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []
A1D-CAE	**0.464**	**119.39**	**52.83**	**0.999**	**0.637**	**117.74**	**59.01**	**1.000**	**0.713**	**118.18**	**55.53**	**0.999**	**0.671**	**116.10**	**51.91**	**0.999**	**0.637**	**114.26**	**55.05**	**1.000**	**0.608**	**118.01**	**53.22**	**0.999**
PCA	0.674	115.75	49.19	0.998	1.049	110.12	51.40	0.999	1.111	109.45	46.81	0.997	0.899	114.07	49.88	0.998	0.905	113.60	54.39	1.000	1.034	113.95	49.16	0.998
NLPCA	0.668	115.88	49.32	0.998	1.143	110.03	51.31	0.999	1.148	109.58	46.94	0.998	0.940	113.07	48.88	0.998	1.110	111.02	51.80	0.999	0.886	114.97	50.17	0.998
SSCNet	4.351	97.43	30.87	0.905	6.192	93.95	35.22	0.950	5.280	95.54	32.89	0.937	3.303	95.84	31.66	0.920	5.971	91.36	32.15	0.926	4.395	96.38	31.58	0.928
SpatialNet	7.924	93.90	27.33	0.810	15.713	89.26	30.53	0.845	9.944	92.77	30.13	0.869	7.807	92.50	28.32	0.839	16.997	87.60	28.38	0.803	12.630	91.11	26.32	0.749
3D-CAE	9.727	90.64	24.08	0.688	12.803	89.66	30.94	0.797	9.742	91.21	28.57	0.777	6.158	91.80	27.61	0.738	13.774	87.85	28.64	0.724	8.998	90.68	25.89	0.745

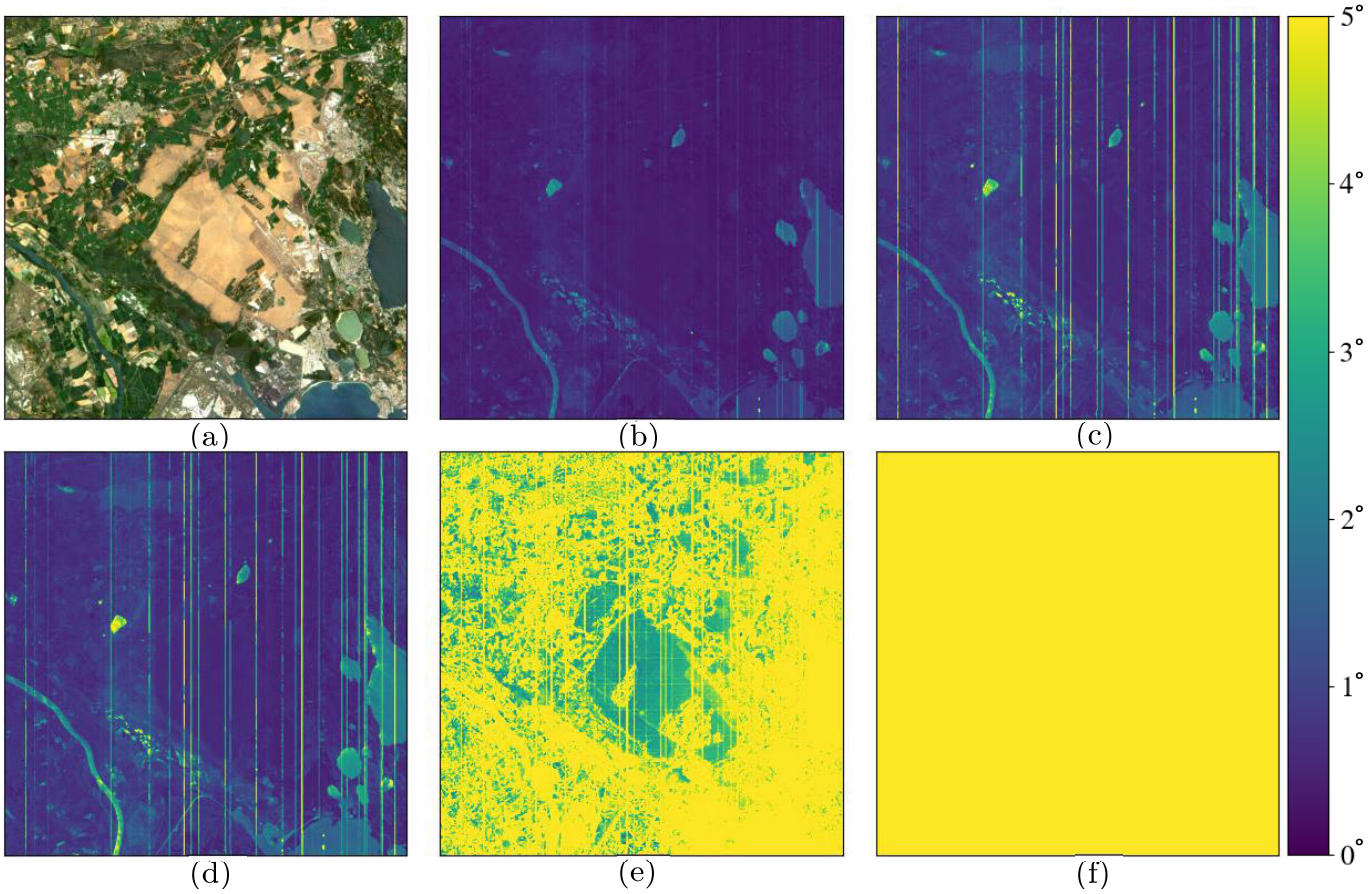


Fig. 7. RGB composition of (a) France test dataset from the PRISMA mission and the error maps based on the SA metric for (b) A1D-CAE and the reference methods (c) PCA, (d) NLPCA, (e) SSCNet, and (f) SpatialNet for the compression rates $c_R = 4$.

For a detailed investigation, we chose the test dataset from France, as shown in Fig. 5(b), because it contains a large number of different spectral signatures. The scene contains bodies of water of different colors and different spectral signatures, which is another interesting aspect. Since the SA is calculated for each pixel, this metric can be used to create an error map to visualize the reconstruction error. Fig. 7 shows the individual SA error maps of the compression methods for the France dataset. To allow a visual comparison, the scale for color coding is fixed from 0° to 5° for all methods. Fig. 7 confirms the SA metric results from Table I and shows that the SSCNet and the SpatialNet have high reconstruction errors. The influence of spatial compression, which includes tiling of the hyperspectral dataset as a preprocessing step, can be seen in the error map of the SSCNet by a tiling pattern, as shown in Fig. 7(e).

All methods have in common that the spectral signatures of water bodies produce the highest reconstruction error. The darker the color of the water body, the higher the reconstruction error. For spectral signatures of dark water bodies, the intensity of the reflected signal is very small, and therefore, the reconstruction error has a greater impact. Other materials and their spectral signatures, such as vegetation, agricultural fields, and urban and industrial areas, are reconstructed with smaller errors. The vertical lines of the PCA and NLPCA error maps could indicate sensor noise from detector elements and significantly affect the reconstruction accuracy of the two methods. The influence on A1D-CAE is significantly smaller, suggesting that the method can also respond to such influences.

Looking at the PSNR metric, the reconstruction accuracy of the A1D-CAE reaches values above 50 dB and the SNR values above 110 dB applied to the hyperspectral data for all test datasets used. With the exception of the USA dataset, the difference in terms of SNR and PSNR is 2–8 dB between the A1D-CAE and the PCA and NLPCA. In a direct comparison, the SNR and PSNR of the A1D-CAE compared to SSCNet

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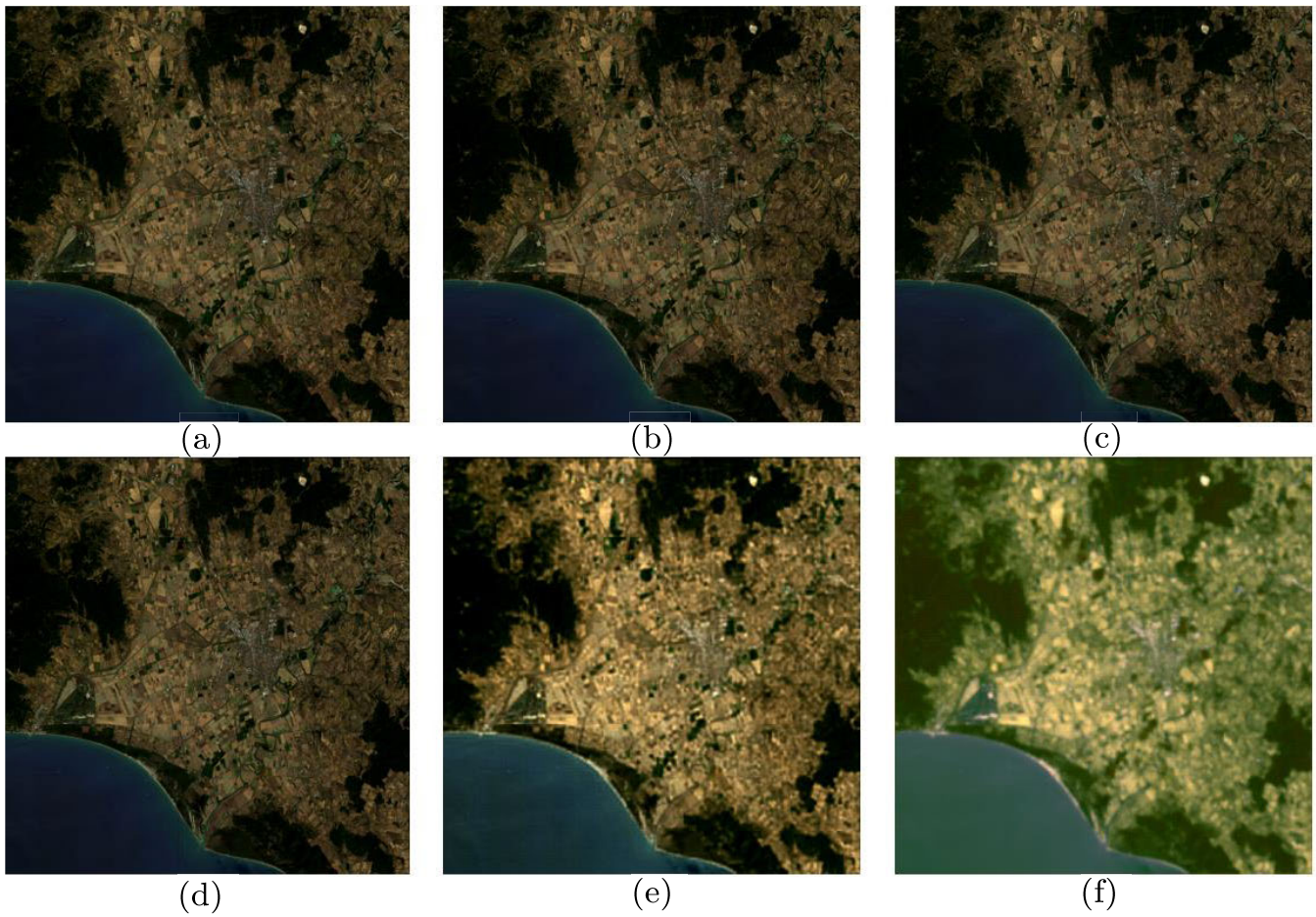


Fig. 8. RGB composition of (a) Italian test dataset from the PRISMA mission and the influence of the reconstructions for (b) A1D-CAE and the reference methods (c) PCA, (d) NLPCA, (e) SSCNet, and (f) SpatialNet for $c_R = 4$ on the edges, textures, and colors in the scene.

TABLE II

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR EXPERIMENT 1 WITH COMPRESSION RATE $c_R = 8$ ON DIFFERENT PRISMA DATASETS BASED ON THE SA, SNR, PSNR, AND SSIM METRICS

$c_R = 8$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []
A1D-CAE	**0.856**	**114.05**	**47.49**	**0.998**	**0.900**	**114.75**	**56.03**	**1.000**	**1.133**	**113.42**	**50.77**	**0.999**	**1.215**	**111.33**	**47.14**	**0.997**	**1.114**	**110.77**	**51.55**	**0.999**	**0.966**	**114.19**	**49.39**	**0.998**
PCA	1.459	109.28	42.72	0.993	2.149	106.39	47.67	0.997	2.133	106.45	43.87	0.995	1.590	109.21	45.02	0.996	1.798	106.32	47.11	0.998	2.283	107.30	42.50	0.993
NLPCA	0.927	113.39	46.82	0.996	1.586	108.25	49.52	0.998	1.588	108.02	45.38	0.997	1.345	109.89	45.70	0.997	1.640	107.21	48.00	0.999	1.280	111.59	46.79	0.997
SSCNet	4.423	97.24	30.68	0.905	6.188	93.80	35.07	0.949	5.336	95.37	32.73	0.935	3.396	95.67	31.48	0.918	6.006	91.19	31.97	0.924	4.510	96.16	31.36	0.926
SpatialNet	8.486	93.44	26.88	0.786	19.844	87.49	28.77	0.777	12.229	91.68	29.03	0.824	9.235	91.58	27.40	0.813	21.813	86.28	27.06	0.731	16.135	89.62	24.82	0.692
3D-CAE	9.711	91.52	24.96	0.682	14.009	88.36	29.64	0.760	9.928	91.70	29.06	0.768	6.216	92.85	28.66	0.737	15.200	86.74	27.53	0.685	8.565	91.96	27.17	0.733

TABLE III

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR EXPERIMENT 1 WITH COMPRESSION RATE $c_R = 16$ ON DIFFERENT PRISMA DATASETS BASED ON THE SA, SNR, PSNR, AND SSIM METRICS

$c_R = 16$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []
A1D-CAE	**1.489**	**109.25**	**42.69**	**0.993**	**1.795**	**108.28**	**49.55**	**0.998**	**1.995**	**107.58**	**44.94**	**0.998**	**1.844**	**107.40**	**43.22**	**0.995**	**2.166**	**105.26**	**46.05**	**0.997**	**1.919**	**108.24**	**43.45**	**0.995**
PCA	2.775	103.77	37.21	0.983	4.492	100.87	42.15	0.992	3.975	101.90	39.26	0.988	3.382	102.77	38.59	0.986	4.966	99.62	40.41	0.991	3.617	103.04	38.25	0.983
NLPCA	1.964	106.54	39.98	0.987	2.829	104.16	45.44	0.996	2.822	104.16	41.52	0.992	2.739	104.49	40.31	0.992	2.718	102.63	43.42	0.996	2.335	106.57	41.78	0.993
SSCNet	4.582	96.96	30.39	0.899	6.340	93.60	34.87	0.947	5.451	95.16	32.52	0.933	3.445	95.49	31.30	0.916	6.113	91.04	31.82	0.922	4.552	95.93	31.14	0.923
SpatialNet	8.504	93.37	26.80	0.779	17.976	88.19	29.47	0.796	11.277	92.01	29.37	0.837	8.463	91.84	27.65	0.814	19.435	86.63	27.42	0.676	15.662	89.73	24.93	0.685
3D-CAE	10.887	90.68	24.12	0.642	15.648	88.64	29.91	0.694	10.814	91.48	28.83	0.727	5.777	92.77	28.59	0.737	14.942	87.23	28.01	0.638	8.989	91.33	26.53	0.706

SpatialNet and 3D-CAE are more than 20 dB higher for each individual test dataset.

The SSIM value for the A1D-CAE ranges from 0.99 to 1.0 for all test datasets. The PCA and NLPCA also achieve similarly good results. This result means that the structures, such as edges, textures, and colors, show almost no distortion at all. The SSCNet and the SpatialNet produce a significantly higher distortion in the structures. This interesting effect could be due to the combination of spatial and spectral features, which likely leads to spectral mixtures that are not adequately unmixed during the reconstruction. Fig. 8 shows the RGB composition of the reconstructed test dataset Italy shown in Fig. 5(c). The visual impression confirms the results of the SSIM metric. The reconstructed dataset of the A1D-CAE retains the structures and colors of the test dataset. On the other hand, the reconstructions of the SSCNet and SpatialNet

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methods show smoothing effects on edges, and the image appears blurred. Furthermore, the colors also deviate from the test dataset.

Looking at the reconstruction accuracy for the compression rate $c_R = 8$ from Table II, the reconstruction error increases for all methods, as expected. It should be noted that the SA values for the A1D-CAE fluctuate around 1° for the different test datasets, thus achieving the highest reconstruction accuracy compared to the reference methods. The reconstruction error for PCA is nearly double compared to A1D-CAE and has increased more rapidly than with other methods. The error of the SSCNet method has remained almost constant.

For the PSNR and SNR metrics, the A1D-CAE scores are between 46 and 56 dB. The difference between A1D-CAE and the other methods is of the same order of magnitude as for the compression rate $c_R = 4$. A1D-CAE's SSIM metric values range from 0.997 to 1.0, which fully reconstructs the structural information. The SSIM values of the PCA, NLPCA, SSCNet, and 3D-CAE have also remained almost constant. Only the SSIM value for SpatialNet has decreased to a greater extent with an increasing compression rate.

Table III shows the reconstruction accuracy for the compression rate $c_R = 16$. The SA values for the A1D-CAE are between 1.4° and 2.1° and, thus, have a low reconstruction error. The reconstruction error of the PCA with respect to the SA metric is twice as large as that of the A1D-CAE. The error difference between NLPCA and A1D-CAE has also doubled. The reconstruction error of SSCNet, SpatialNet, and 3D-CAE has remained almost constant compared to the compression rate $c_R = 8$. The A1D-CAE achieves values between 43 and 49 dB for the PSNR and thus partly higher PSNR values than the state-of-the-art methods at the compression rate $c_R = 8$. The SNR value of the A1D-CAE is above 105 dB for all test datasets, indicating a higher reconstruction accuracy than for the SSCNet, SpatialNet, and 3D-CAE methods at a compression rate of $c_R = 4$.

Fig. 9 shows the error map for the compression rates $c_R = 4$, $c_R = 8$, and $c_R = 16$ of the A1D-CAE for the France test dataset. It can be observed that the reconstruction error generally increases with increasing compression rate. However, the reconstruction error for spectral signatures of vegetation, agricultural fields, and urban and industrial areas remains below the 3° threshold mentioned in [38] for all compression rates. Higher reconstruction errors occur with bodies of water, shadowed areas, and noisy bands.

The test dataset Switzerland [see Fig. 5(e)] shows other interesting effects. The spectral signature of the snow on the mountain was not part of the training dataset. Looking at the SA error map of Fig. 10, the reconstruction error is not significantly higher in snow-covered areas even though the spectral signature is unknown and shows no similarity to other materials. It is noteworthy that the highest reconstruction error comes from heavily shadowed areas. Shaded areas have characteristics similar to very dark bodies of water because their reflected signal has a relatively low amplitude. Thus, the spectrum is more difficult to reconstruct because it contains less information, and the reconstruction error has a greater influence. Another assumption is that the dynamic range has been shifted in favor of the snow spectra, further reducing the intensity of the spectral signature of the shadow areas.

B. Evaluation Experiment 2

In Section III-C2, the A1D-CAE architecture is extended to the CompNext1D with the second compression stage based on image statistics, as shown in Fig. 1. The state-of-the-art methods were upgraded in the same way for comparability. For the results in Table IV, the A1D-CAE with a compression rate of $c_R = 4$ was used as the base model and extended by a second stage to a compression rate of $c_R \approx 16$. Remarkably, the reconstruction error related to the SA metric for the CompNext1D is only slightly higher than for the A1D-CAE although the compression rate is four times higher. The additional loss of information caused by the k-means quantization with $n_{\text{cluster}} = 250$ of the second compression stage can be almost eliminated by the generalization capability of the A1D-CAE.

Thus, the difference between CompNext1D and A1D-CAE based on the SA metric is in the range of 0.01°–0.019° for all test datasets. Considering the SNR and PSNR metrics, the difference between CompNext1D and A1D-CAE is in the range of 0.5–5.1 dB. There is no difference in the SSIM metrics. Compared to extended state-of-the-art methods, the reconstruction error for CompNext1D is lower than for the basic models.

Table V shows the results for the compression rate $c_R \approx 32$ with the basic model A1D-CAE $c_R = 8$ and a cluster number of $n_{\text{cluster}} = 250$. Table VI shows the results for the compression rate $c_R \approx 64$ with the basic model A1D-CAE $c_R = 8$ and a cluster number of $n_{\text{cluster}} = 18$. It is noteworthy that the reconstruction error for CompNext1D with $n_{\text{cluster}} = 250$ is slightly higher compared to the A1D-CAE base model $c_R = 8$, and thus, the results of CompNext1D are comparable to the base model $c_R = 4$ because of identical values for n_{cluster}. The SA values of the CompNext1D are 0.007°–0.013° higher than the base model. Furthermore, the reconstruction accuracy based on the PSNR and SNR metrics is only 0.13–2.77 dB lower. The SSIM values remain the same. The differences in the reconstruction accuracy of CompNext1D compared to the state-of-the-art methods are almost similar to those of the corresponding base models.

Various tests have shown that the parameter $n_{\text{cluster}} = 18$ doubles the compression rate for the same base model. The reconstruction error increases on average by 0.863°, the PSNR and SNR decrease by 6.46 dB, and the SSIM metric decreases by 0.006 compared to the base model. The small increase in terms of the SA metric and the comparatively larger loss in terms of the PSNR and SNR metrics can be explained by the fact that the information loss tends to cause high-frequency interference, such as ripple and distortion on the spectral signature, rather than smoothing it. This type of error has a greater impact on the PSNR and SNR metrics than on the SA metric.

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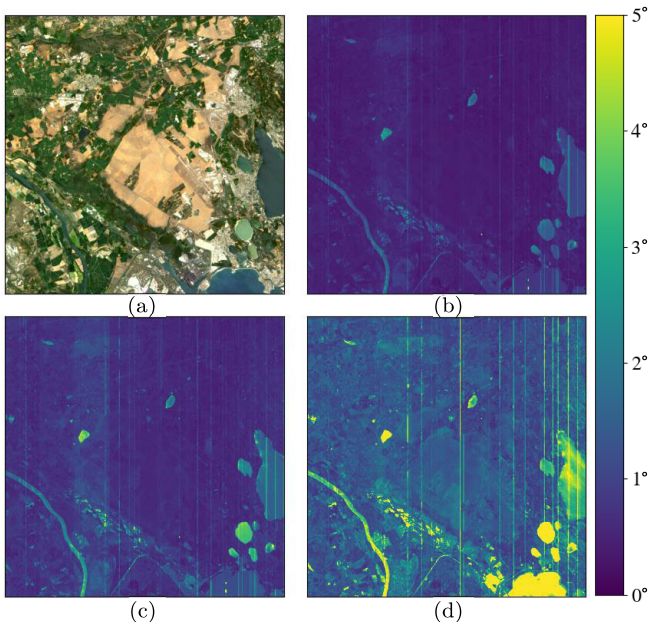


Fig. 9. RGB composition of (a) France test dataset from the PRISMA mission and the error maps based on the SA metric for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$ of the A1D-CAE.

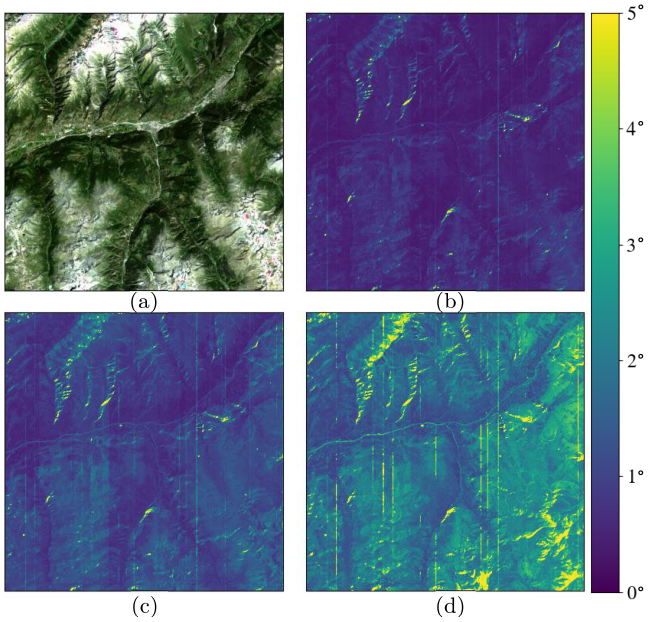


Fig. 10. RGB composition of (a) Switzerland test dataset from the PRISMA mission and the error maps based on the SA metric for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$ of the A1D-CAE.

TABLE IV

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR THE COMPNEXT1D FOR THE EXPERIMENT 2 WITH COMPRESSION RATE $c_R = 4$ OF THE A1D-CAE BASE MODEL AND $n_{\text{CLUSTER}} = 250$ ON UNKNOWN PRISMA DATASETS. THE METRICS SA, SNR, PSNR, AND SSIM ARE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

base model $c_R = 4$ $c_R \approx 16$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM
CompNext1D	**0,472**	**118,86**	**52,30**	**0,999**	**0,649**	**115,62**	**56,90**	**1,000**	**0,723**	**117,03**	**54,38**	**0,999**	**0,681**	**113,90**	**49,71**	**0,999**	**0,655**	**109,09**	**49,88**	**1,000**	**0,617**	**117,26**	**52,46**	**0,999**
PCA	0,686	112,50	45,93	0,998	1,165	107,51	48,79	0,998	1,200	107,63	44,99	0,997	0,941	111,84	47,65	0,997	1,021	105,19	45,97	0,999	1,048	112,98	48,19	0,998
NLPCA	0,687	115,55	48,98	0,998	1,190	109,04	50,31	0,999	1,174	109,43	46,79	0,997	0,969	112,48	48,30	0,998	1,146	109,79	50,58	0,999	0,912	114,52	49,72	0,998
SSCNet	4,582	96,96	30,39	0,899	6,340	93,60	34,87	0,947	5,451	95,16	32,52	0,933	3,445	95,49	31,30	0,916	6,113	91,04	31,82	0,922	4,552	95,93	31,14	0,923
SpatialNet	8,504	93,37	26,80	0,779	17,976	88,19	29,47	0,796	11,277	92,01	29,37	0,837	8,463	91,84	27,65	0,814	19,435	86,63	27,42	0,751	15,662	89,73	24,93	0,685
3D-CAE	9.795	90.60	24.04	0.686	12.839	89.64	30.92	0.797	9.748	91.20	28.56	0.777	6.155	91.75	27.57	0.738	15.147	86.78	27.57	0.711	9.022	90.69	25.89	0.745

Table VII shows the results for the compression rate $c_R \approx 100$ with the basic A1D-CAE model $c_R = 16$ and a cluster number of $n_{\text{cluster}} = 35$. The compression rate was increased by a factor of 6.25 compared to the base model. At the same time, the mean reconstruction error based on the SA metric increased by 0.476°. Furthermore, the reconstruction

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TABLE V

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR THE COMPNEXT1D FOR THE EXPERIMENT 2 WITH COMPRESSION RATE $c_R = 8$ OF THE A1D-CAE BASE MODEL AND $n_{\text{CLUSTER}} = 250$ ON UNKNOWN PRISMA DATASETS. THE METRICS SA, SNR, PSNR, AND SSIM ARE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

base model $c_R = 8$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
$c_R \approx 32$	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM
	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]
CompNext1D	**0.863**	**113.92**	**47.35**	**0.998**	**0.912**	**113.79**	**55.07**	**1.000**	**1.146**	**113.02**	**50.38**	**0.999**	**1.223**	**109.61**	**45.43**	**0.997**	**1.125**	**107.99**	**48.78**	**0.999**	**0.977**	**113.73**	**48.94**	**0.998**
PCA	1.317	109.46	42.90	0.995	2.007	106.02	47.30	0.997	2.165	105.73	43.09	0.994	1.480	107.55	43.36	0.995	2.096	103.08	43.86	0.997	1.821	109.09	44.30	0.995
NLPCA	0.935	113.24	46.68	0.996	1.602	108.11	49.38	0.998	1.600	107.95	45.30	0.996	1.354	109.47	45.28	0.996	1.657	107.00	47.78	0.998	1.291	111.44	46.64	0.997
SSCNet	4.357	97.43	30.86	0.905	6.626	93.63	34.90	0.945	5.281	95.53	32.89	0.937	3.300	95.84	31.65	0.920	5.966	91.36	32.15	0.926	4.392	96.38	31.58	0.928
SpatialNet	9.078	92.77	26.21	0.780	21.515	86.89	28.16	0.758	13.321	91.11	28.46	0.816	10.290	90.98	26.80	0.800	22.659	85.83	26.62	0.718	18.189	88.77	23.98	0.672
3D-CAE	10.102	90.38	23.82	0.667	14.001	88.36	29.64	0.760	9.929	91.70	29.05	0.768	6.216	92.69	28.51	0.737	15.443	85.55	26.33	0.682	8.563	91.96	27.16	0.733

TABLE VI

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR THE COMPNEXT1D FOR THE EXPERIMENT 2 WITH COMPRESSION RATE $c_R = 8$ OF THE A1D-CAE BASE MODEL AND $n_{\text{CLUSTER}} = 18$ ON UNKNOWN PRISMA DATASETS. THE METRICS SA, SNR, PSNR, AND SSIM ARE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

base model $c_R = 8$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
$c_R \approx 64$	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM
	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]
CompNext1D	**1.565**	**107.07**	**40.51**	**0.988**	**1.764**	**105.72**	**46.99**	**0.997**	**1.874**	**107.02**	**44.37**	**0.994**	**1.658**	**106.99**	**42.80**	**0.993**	**2.236**	**103.78**	**44.56**	**0.995**	**1.755**	**107.15**	**42.36**	**0.992**
PCA	1.648	107.24	40.68	0.989	2.776	102.55	43.83	0.993	2.491	103.74	41.10	0.991	1.808	106.56	42.37	0.992	3.185	92.89	33.67	0.984	2.387	105.36	40.56	0.987
NLPCA	1.680	105.27	38.71	0.983	3.025	101.26	42.53	0.991	2.797	101.67	39.03	0.983	2.275	102.55	38.36	0.980	3.302	99.95	40.73	0.988	2.414	103.64	38.84	0.982
SSCNet	6.130	93.64	27.08	0.867	12.803	82.39	23.67	0.853	12.327	86.81	24.17	0.863	8.172	90.95	26.77	0.875	11.938	80.12	20.90	0.778	16.003	86.93	22.13	0.779
SpatialNet	11.099	90.26	23.69	0.703	19.805	87.44	28.71	0.776	13.144	90.32	27.68	0.782	12.112	87.74	23.56	0.740	22.379	83.67	24.46	0.718	18.783	85.48	20.69	0.639
3D-CAE	10.657	89.28	22.71	0.653	13.984	88.36	29.64	0.761	9.947	91.67	29.02	0.768	6.227	92.54	28.35	0.737	15.230	86.60	27.38	0.685	8.557	91.93	27.14	0.733

TABLE VII

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR THE COMPNEXT1D FOR THE EXPERIMENT 2 WITH COMPRESSION RATE $c_R = 16$ OF THE A1D-CAE BASE MODEL AND $n_{\text{CLUSTER}} = 35$ ON UNKNOWN PRISMA DATASETS. THE METRICS SA, SNR, PSNR, AND SSIM ARE USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

base model $c_R = 16$	Brazil 2020.08.06				France 2020.08.02				Italy 2020.08.01				USA 2021.03.01				Switzerland 2020.08.07				Myanmar 2021.05.08			
$c_R \approx 100$	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM	SA	SNR	PSNR	SSIM
	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]	[°]	[dB]	[dB]	[]
CompNext1D	**1.737**	**107.39**	**40.83**	**0.989**	**2.489**	**101.93**	**43.21**	**0.993**	**2.521**	**103.92**	**41.28**	**0.992**	**1.983**	**104.99**	**40.80**	**0.991**	**2.849**	**101.46**	**42.24**	**0.993**	**2.132**	**106.20**	**41.41**	**0.992**
PCA	2.500	104.54	37.98	0.984	3.325	102.80	44.08	0.994	3.424	102.81	40.17	0.990	3.181	102.33	38.14	0.985	3.796	100.92	41.70	0.993	3.453	103.32	38.52	0.986
NLPCA	2.080	105.29	38.73	0.982	3.078	102.45	43.72	0.993	2.996	103.03	40.39	0.989	2.837	103.59	39.41	0.992	2.963	101.31	42.09	0.993	2.449	105.68	40.89	0.990
SSCNet	6.316	93.45	26.89	0.864	13.035	81.57	22.85	0.841	12.402	86.69	24.05	0.862	9.223	90.00	25.82	0.867	11.469	81.13	21.92	0.801	14.165	88.20	23.41	0.804
SpatialNet	21.677	82.88	16.31	0.519	20.333	86.70	27.97	0.741	15.780	88.41	25.76	0.706	14.838	85.90	21.71	0.660	21.456	83.51	24.29	0.712	25.191	81.43	16.63	0.543
3D-CAE	12.594	87.98	21.41	0.581	16.366	87.98	29.25	0.708	10.806	91.47	28.82	0.729	5.933	91.47	27.29	0.735	17.056	85.39	26.18	0.625	9.268	90.65	25.86	0.716

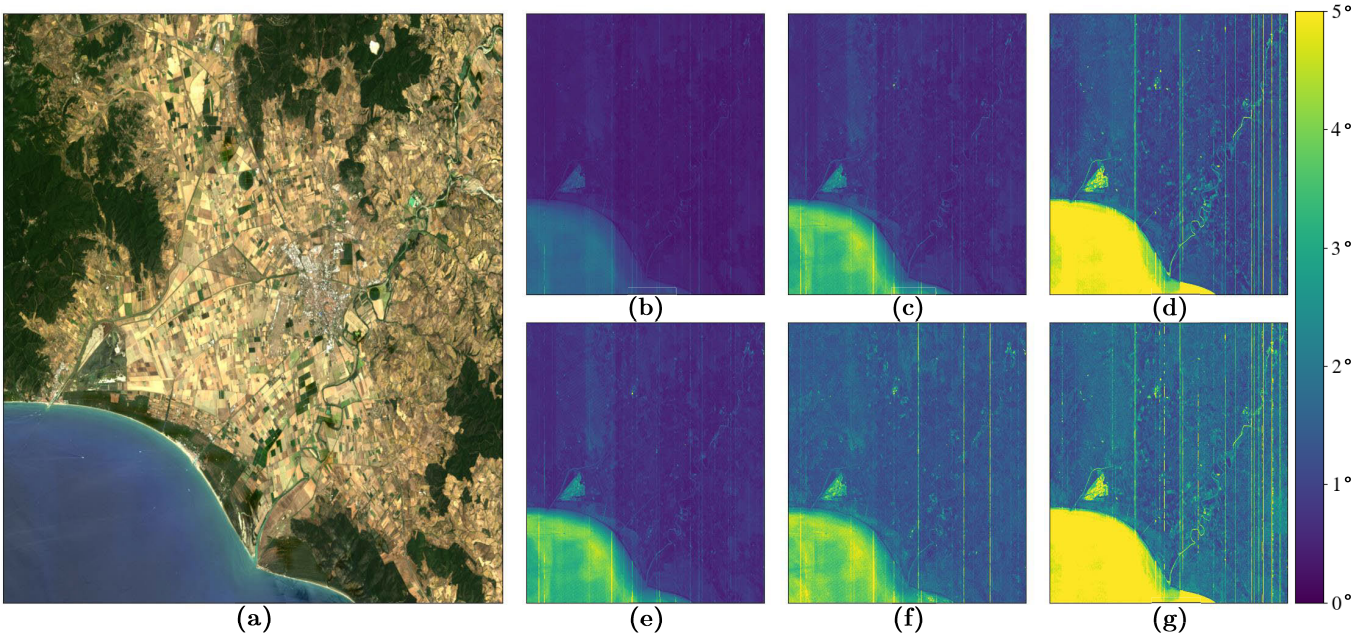


Fig. 11. RGB composition of (a) Italy test dataset from the PRISMA mission and the error maps based on the SA metric of the A1D-CAE for Experiment 1 and Experiment 2. The images in the top row show the reconstruction error of the A1D-CAE base model for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$. The images in the bottom row show the error map of the CompNext1D for the compression rates (e) $c_R \approx 32$, (f) $c_R \approx 64$, and (g) $c_R \approx 100$.

accuracy decreased by 3.49 dB on average, and the SSIM metric decreased by 0.004. It is noteworthy that, even with a compression rate of $c_R \approx 100$, the SA value averaged over all datasets reaches 2.55°. The CompNext1D achieves an average SNR of 103.86 dB, a PSNR of 40.85 dB, and an SSIM of 0.990.

Fig. 11 shows the error map for the base A1D-CAE models and CompNext1D for the different compression rates. The error maps show that the CompNext1D reconstruction errors are very close to the base models when n_{cluster} is set to high values. As the value of n_{cluster} is reduced, the reconstruction accuracy decreases. The error maps in Fig. 11 confirm the

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TABLE VIII
EVALUATION OF THE INFLUENCE OF THE LOSSY COMPRESSION METHODS FOR THE COMPRESSION RATE $c_R = 4$ OF COMPNEXT1D ON THE RECONSTRUCTION ERROR

$c_R = 4$	SA [°]	SNR [dB]	PSNR [dB]	SSIM []
CompNext1D	0.64	115.62	56.90	1.000
A1D-CAE	0.63	115.74	59.01	1.000
k-means quantization	0.12	115.01	63.39	0.999
Data typ conversion	0.11	115.22	63.60	0.999

previous finding that the highest reconstruction errors occur in water bodies and noisy bands. In particular, the reconstruction accuracy for water bodies exceeds the limit of 3° from [38] for the compression rates $c_R = 16$ of the base model and $c_R \approx 64$ and $c_R \approx 100$ for CompNext1D. However, other spectral signatures, such as vegetation and agricultural land, remain below 3° for all compression rates presented.

To further evaluate the accuracy of the spectral reconstruction, two spectral signatures were extracted from the France dataset. The selected spectral signatures are not representative of the entire dataset but are intended to contribute to the understanding and visualizing the reconstruction error. For the compression rates $c_R = 4$ (orange, dotted), $c_R = 8$ (green, dashdot), and $c_R = 16$ (red, dotted), the A1D-CAE was used. For the compression rate $c_R \approx 32$ (purple, densely dashdotdotted) and $c_R \approx 64$ (brown, dashdotdotted), the A1D-CAE $c_R = 8$ was used as the base model in the CompNext1D architecture, and for the k-means quantization, $n_{\text{cluster}} = 250$ and $n_{\text{cluster}} = 35$. For the compression rate $c_R \approx 100$ (pink, dashdotdotted), $c_R = 16$ was used as the basis in the CompNext1D model of the A1D-CAE and $n_{\text{cluster}} = 250$ for the k-means quantization. We compare the original spectral signature (blue, solid) with the different reconstructions. Fig. 12 shows the spectral signature of the class vegetation. The reconstruction, except for the compression rate $c_R \approx 100$, shows a high agreement of the spectral shape. All reconstructions can recover the red edge feature in the 680–720-nm wavelength range with a very small reconstruction error. Fig. 13 compares the spectral signature of an urban environment and the reconstructions for the different reconstruction factors. The general spectral shape is recovered for all compression rates. For the features themselves, the reconstruction ($c_R \approx 64$ and $c_R \approx 100$) has an offset and, therefore, a small deviation from the original spectrum.

At the end of this section, we want to evaluate the reconstruction error of the individual components, such as A1D-CAE, k-means quantization, and data type conversion of the multistage compression process. The reconstruction error is determined for each process individually and compared to CompNext1D, as shown in Table VIII. The A1D-CAE is the largest contributor to the reconstruction error of the CompNext1D model. The k-means quantization and the data type conversion have a comparatively low reconstruction error, as shown in Table VIII. Measured by the SNR and SSIM metrics, the A1D-CAE has a higher influence on the reconstruction accuracy. This pattern is repeated for higher compression rates. Therefore, k-means quantization, data type conversion, and Huffman coding are used after A1D-CAE to increase the compression rate at a small tradeoff based on image statistics.

C. Evaluation Experiment 3

Table IX shows the results using the A1D-CAE base model for the compression rates $c_R = 4$, $c_R = 8$, and $c_R = 16$ on hyperspectral data from different and unknown sensors. Since the state-of-the-art methods cannot be applied to the test data from other sensors, no results of the state-of-the-art methods are available for this experiment.

The A1C-CAE with a compression rate of $c_R = 4$ achieves a reconstruction accuracy of 1.56° for the satellite-borne DESIS test dataset. The SNR value is 114.88 dB, and the PSNR value is 41.70 dB with an SSIM value of 0.990. These results indicate a high level of reconstruction accuracy. As the compression rate increases, the reconstruction error also increases, so that, for the compression rate $c_R = 16$, an SA value of 5.26° is calculated, which exceeds the proclaimed limit of 3° [38]. Fig. 14 shows the A1D-CAE error maps of the DESIS test dataset for the compression rates $c_R = 4$, $c_R = 8$, and $c_R = 16$. The scene consists mainly of vegetation, agricultural fields, urban areas, and water bodies. It is noteworthy that, in this test dataset, water bodies are reconstructed with a similarly low level of error as the other spectral signatures that occur. The red-marked area in the scene shows a lake. It can be seen that the reconstruction error is even lower than the surrounding vegetation or agricultural fields.

For the airborne Greding test dataset, the reconstruction accuracy of the A1D-CAE is 2.22° and 103.74 dB for the SNR, 40.27 dB for the PSRN, and 0.990 for the SSIM. If the compression rate is increased to $c_R = 8$, the reconstruction accuracy drops to 4.7°, and the SNR and PSNR values drop by 5.5 dB. At $c_R = 16$, the SA value deteriorates to 9.31° and the SSIM to 0.85, no longer representing a sufficient reconstruction. Fig. 15 shows the error map of the Greding test dataset for compression rates $c_R = 4$, $c_R = 8$, and $c_R = 16$. The urban scene mainly contains spectral signatures of vegetation, roads, houses, and fields. It can be seen that the highest reconstruction errors are assigned to the shaded areas. Looking at the areas inside the red dashed boxes, shadow areas are cast west of the houses. These areas have a significantly higher reconstruction error than the surrounding buildings, roads, or vegetation. The best reconstructions can be obtained for the classes of streets and houses.

For the test dataset KHH, measured with a drone-based system, and for $c_R = 4$, a reconstruction accuracy of 1.24° and 47.78 dB for the PSNR and an SSIM value of 1.00 are obtained. Only when the compression rate is quadrupled to $c_R = 16$, the accuracy drops to SA = 5.74°. PSNR = 34.20 dB, and the SSIM value is 0.980 Fig. 16 shows the corresponding RGB composition. The scene mostly shows a barren agricultural field with crops in the northern part and grassy areas in the south. Black frames 10 × 10 cm were placed on the center area to mark positions for the spectral reference measurements and the GPS station. The error map for the compression rate $c_R = 4$ shows a small reconstruction error for the agricultural field and vegetation.

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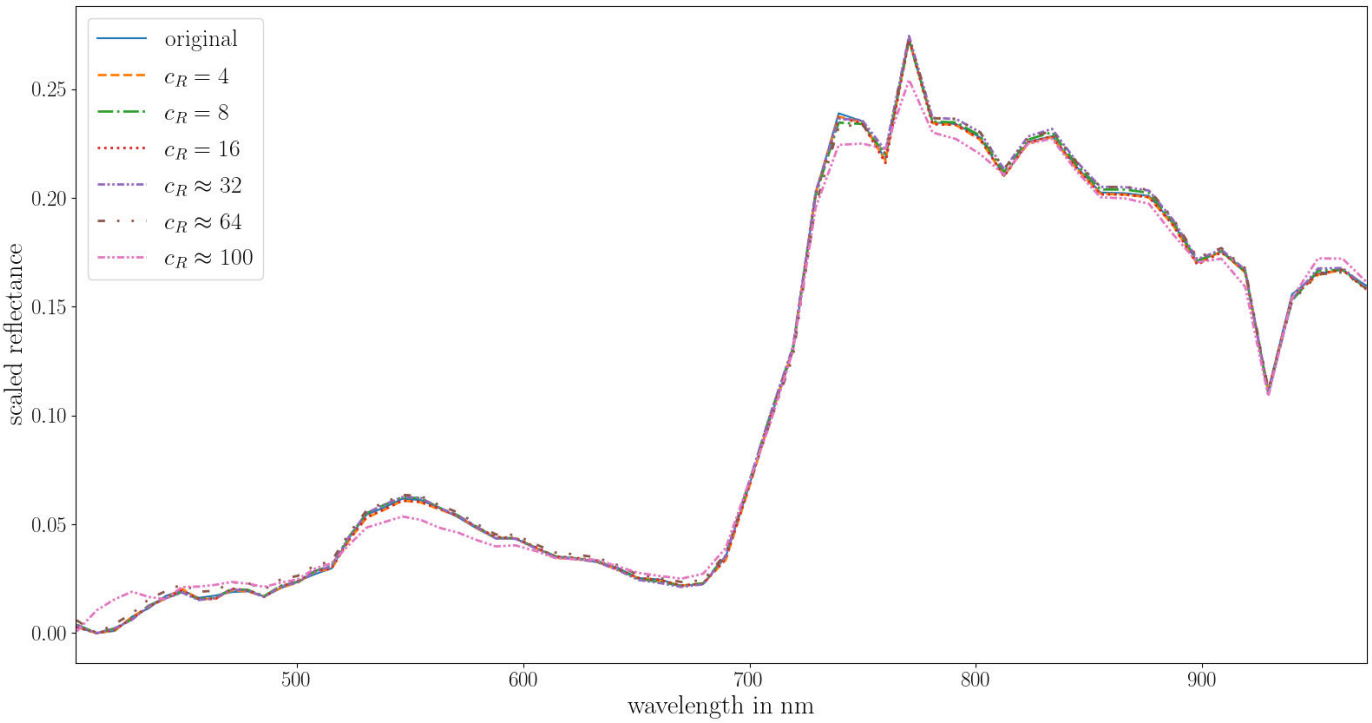


Fig. 12. Comparison between the original spectral signature of the class vegetation and the reconstruction of the A1D-CAE ($c_R = 4$, $c_R = 8$, and $c_R = 16$) and CompNexpt1D ($c_R \approx 32$, $c_R \approx 64$, and $c_R \approx 100$).

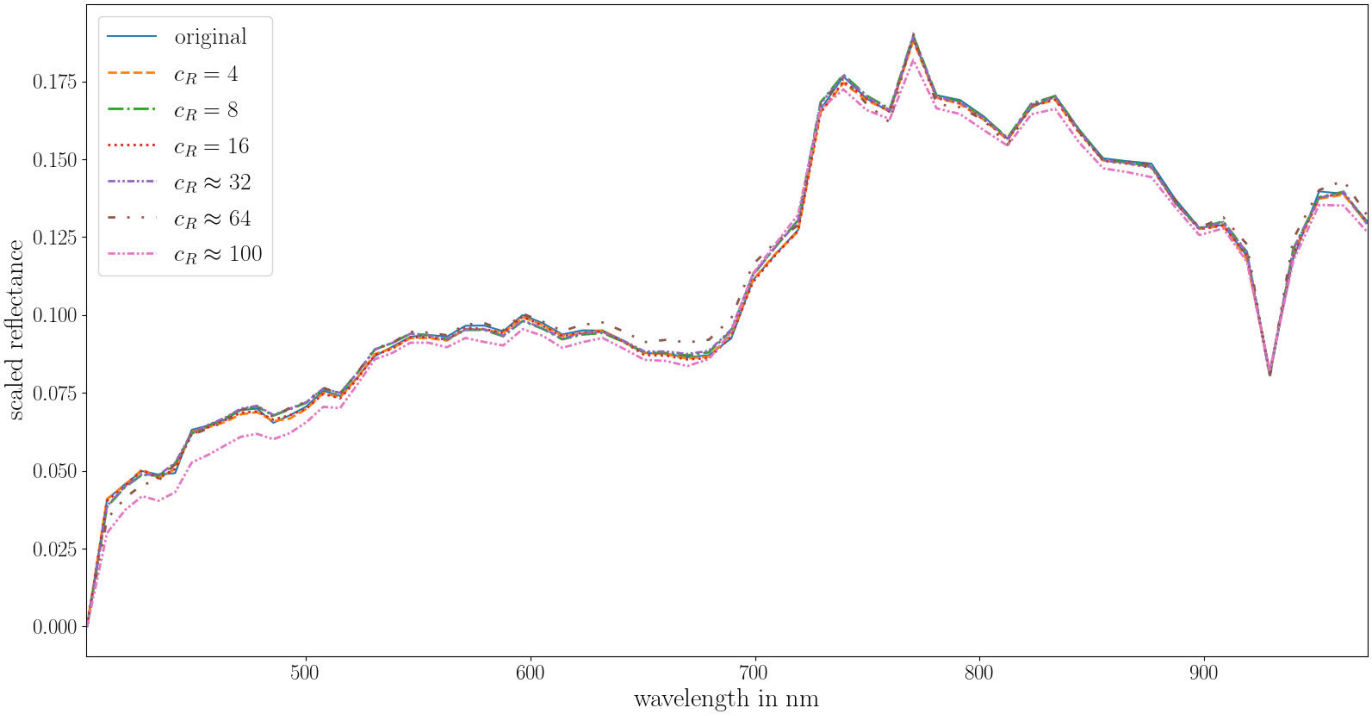


Fig. 13. Comparison between the original spectral signature of the class urban area and the reconstruction of the A1D-CAE ($c_R = 4$, $c_R = 8$, and $c_R = 16$) and CompNexpt1D ($c_R \approx 32$, $c_R \approx 64$, and $c_R \approx 100$).

The reason might be this object's very different material and colors compared to the training dataset. This could be because the material and colors of this object are very different from the training data. For the compression rate of $c_R = 16$, the reconstruction error of each pixel exceeds the SA value of 5°. For the PaviaU benchmark dataset, the A1D-CAE achieves a reconstruction error of SA = 2.71°, PSNR = 39 dB, and an SSIM value of 0.99 for $c_R = 4$. With a compression rate of $c_R = 16$, the compression error increases to 9.78° and 27.29 dB with an SSIM value of 0.90. Fig. 17 shows the RGB composition of the scene and the corresponding error maps for compression rates $c_R = 4$, $c_R = 8$, and

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TABLE IX

EVALUATION OF THE RECONSTRUCTION ACCURACY OF THE A1D-CAE FOR EXPERIMENT 3 FOR THE COMPRESSION RATES $c_R = 4$, $c_R = 8$, AND $c_R = 16$ ON DIFFERENT DATASETS FROM DIFFERENT HYPERSPECTRAL SENSORS BASED ON THE METRICS SA, SNR, PSNR, AND SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

A1D-CAE	DESIS				Greding				KHH				PaviaU			
	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []
$c_R = 4$	**1.56**	**114.88**	**41.70**	**0.99**	**2.22**	**103.74**	**40.27**	**0.99**	**1.24**	**117.87**	**47.78**	**1.00**	**2.71**	**98.61**	**39.00**	**0.99**
$c_R = 8$	3.05	109.13	35.95	0.98	4.70	98.21	34.74	0.97	2.89	109.94	39.85	0.99	4.82	94.16	34.56	0.98
$c_R = 16$	5.27	103.22	30.04	0.93	9.31	87.01	23.53	0.85	5.74	104.29	34.20	0.98	9.78	86.90	27.29	0.90

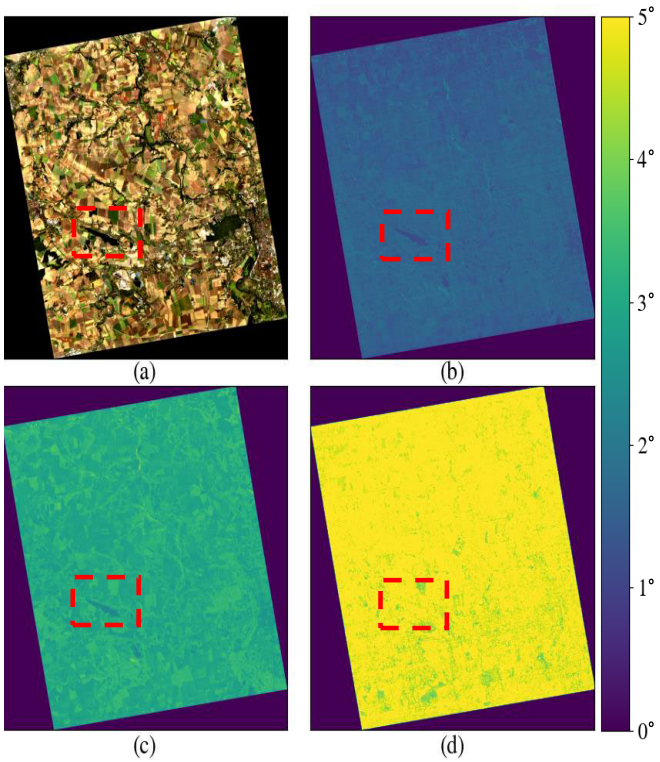


Fig. 14. RGB composition of (a) DESIS test dataset and the error maps based on the SA metric for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$ of the A1D-CAE. The red dashed box marks the position of a lake in the RGB image and the corresponding position in the error maps.

$c_R = 16$. The scene is an urban area with vegetation. A similar effect can be seen in the Greding test dataset, where shaded areas have a significantly higher reconstruction error. This effect is more pronounced in this dataset. Unexpectedly, the A1D-CAE has problems reconstructing vegetation, and the reconstruction error is higher compared to buildings or roads. This effect appears in the PaviaU test dataset for the first time, as the previous experimental and test datasets reconstructed vegetation with a small reconstruction error.

D. Ablation Study

In this ablation study, we aim to understand the impact of different configurations on the hyperspectral compression performance of three selected datasets: Brazil, France, and Switzerland. The configurations considered A1D-CAE as a baseline. The modifications are divided into the following three groups:

1) feature extraction and reconstruction;
2) activation function;
3) loss functions.

For feature extraction, average pooling and maximum pooling are investigated. For feature reconstruction, the upsampling layer is selected. The widely used leaky ReLU, sigmoid, and ELU are examined as activation functions. The SAM metric is used as the loss function instead of the mse. The performance of the models is evaluated in terms of SA, SNR, PSNR, and SSIM. Table X shows the reconstruction accuracy for the feature extraction with average pooling and max pooling, and the feature reconstruction with upsampling. Feature extraction was performed in the baseline using strides within the convolutional layer. The feature reconstruction was relayed in A1D-CAE with a transposed convolution layer. The ablation study shows in Table X that the baseline is achieved for all three datasets and all metrics. The above analysis demonstrates that the A1D-CAE is built with the components for the highest reconstruction accuracy. The results for examining different activation functions show that the GELU activation function used in A1D-CAE produces the lowest reconstruction error. The highest reconstruction error is generated with the sigmoid activation function. The results demonstrate that the GELU activation function in the A1D-CAE architecture produces the highest reconstruction accuracy. Table X compares the SA as a loss function with the mse, which was used for training the A1D-CAE. It can be clearly seen that the reconstruction

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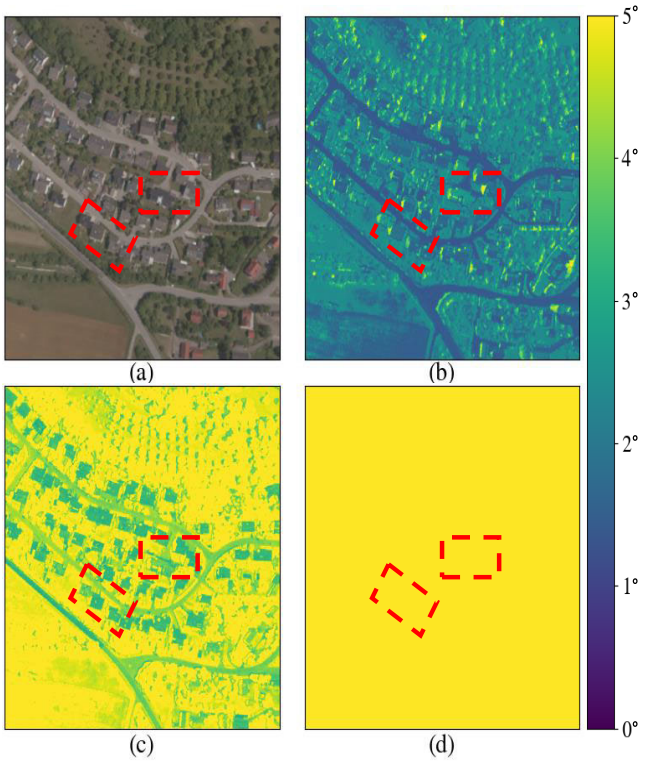


Fig. 15. RGB composition of (a) Greding test dataset and the error maps based on the SA metric for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$ of the A1D-CAE. The red dashed box marks the position of the shadow areas in the RGB image and the corresponding position in the error maps.

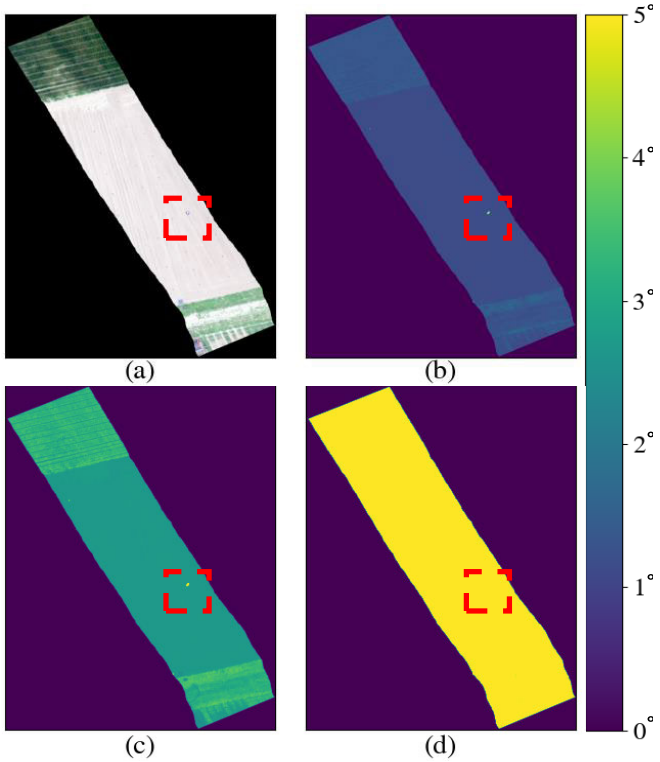


Fig. 16. RGB composition of (a) KHH test dataset and the error maps based on the SA metric for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$ of the A1D-CAE. The red dashed box marks the position of the GPS station in the RGB image and the corresponding position in the error maps.

error is higher for the model that was trained with the SA as a loss function. Handling the training became significantly more complex using SA as a loss, as it is numerically unstable. This is because the SA metric returns an angle between 0 and π, which means that the gradient can become very small in certain areas, making training difficult. Measures such as gradient clipping and LR adjustment had to be considered to converge the model. Thus, the SA metric is not suitable

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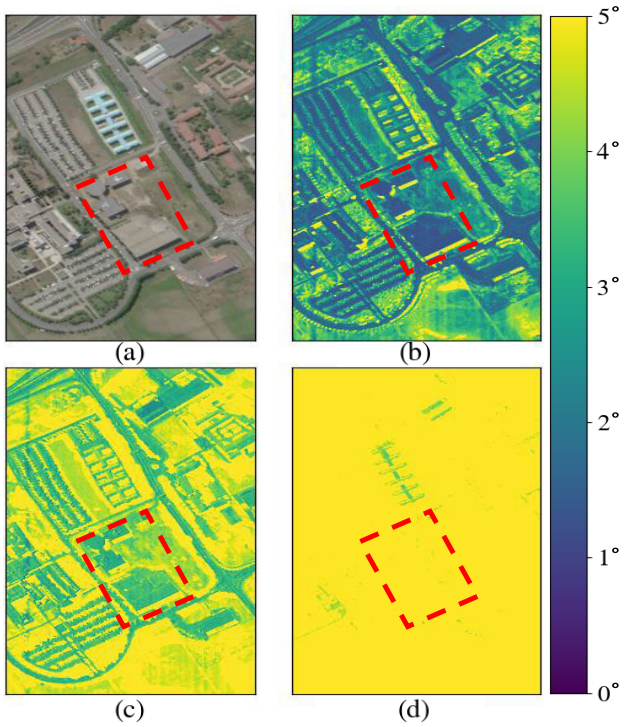


Fig. 17. RGB composition of (a) PaviaU test dataset and the error maps based on the SA metric for the compression rates (b) $c_R = 4$, (c) $c_R = 8$, and (d) $c_R = 16$ of the A1D-CAE. The red dashed box marks the position of the shadow areas in the RGB image and the corresponding position in the error maps.

TABLE X

EVALUATION OF THE RECONSTRUCTION ACCURACY FOR THE ABLATION STUDY WITH COMPRESSION RATE $c_R = 4$ ON DIFFERENT PRISMA DATASETS BASED ON THE SA, SNR, PSNR, AND SSIM METRICS

Model	Brazil 2020.08.06				France 2020.08.02				Switzerland 2020.08.07			
	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []	SA [°]	SNR [dB]	PSNR [dB]	SSIM []
A1D-CAE	**0.464**	**119.39**	**52.83**	**0.999**	**0.637**	**117.74**	**59.01**	**1.000**	**0.637**	**114.26**	**55.05**	**1.000**
average pooling	0.523	118.91	51.23	0.998	0.727	117.34	58.50	0.999	0.749	113.06	54.73	0.999
max pooling	0.550	118.62	51.94	0.998	0.724	112.92	58.82	0.999	0.703	117.66	54.59	0.999
upsampling	0.521	118.73	52.05	0.998	0.723	117.16	58.32	0.999	0.720	111.57	53.23	0.999
leaky-ReLU	0.531	118.51	51.33	0.998	0.738	117.46	58.61	0.999	0.743	113.44	54.10	0.999
Sigmoid	1.458	105.87	40.19	0.991	3.845	100.06	42.22	0.988	3.816	95.04	36.70	0.989
ELU	0.545	118.46	51.78	0.998	0.779	114.80	56.95	0.999	0.774	112.72	54.39	0.999
SA-loss	6.754	88.87	22.31	0.726	9.734	73.01	14.29	0.353	9.470	72.87	13.66	0.304

TABLE XI

EVALUATION OF THE COMPUTATIONAL COMPLEXITY BASED ON THE INPUT SHAPE, NUMBER OF PARAMETERS, GIGA FLOATING POINT OPERATIONS PER SECOND (GFLOPS), AND STORAGE SPACE FOR THE COMPRESSION RATE $c_R = 4$

Model	Input shape ($B \times m \times n \times b$)	Params [Mio.]	FLOPS [GFLOPS]	Storage space [Mb]
A1D-CAE	1, 63	1.74	0.04	6.63
NLPCA	1, 63	0.01	6.32e−6	0.02
SSCNet	1, 63, 64, 64	13.14	6.60	50.14
SpatialNet	1, 63, 1000, 1000	0.37	106.56	1.40
PCA	1, 63	2.09e−3	2.09e−6	0.01
3D-CAE	1, 1, 63, 64, 64	0.97	4.23	3.72

as a loss function. Combining the SA metric with another metric might be helpful to obtain regularization to stabilize training.

E. Computational Complexity

The computational complexity of our proposed A1D-CAE and the reference methods is listed in Table XI. Since the computational complexity depends heavily on the input form, comparing the 1-D methods with the 2-D and 3-D methods is difficult. Table XI shows that 2-D and 3-D methods perform significantly higher FLOPS than 1-D. The number of parameters and the storage space required for the model depend not on the input form but on the depth of the architecture. However, the number of parameters and the memory required for all compression methods are minimal compared to language and vision models with several billion parameters.

F. Computational Time

When evaluating the computational time for inferencing the models, the input form of the data should be considered. The inferencing was carried out with an NVIDIA Tesla V100 GPU. The input data were compressed and reconstructed 1000 times to determine the average computation time. Furthermore, the standard deviation of the series of measurements was calculated. The results of the measurement for the computational

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TABLE XII
COMPARISON OF THE COMPUTATION TIME FOR THE COMPRESSION METHODS FOR $c_R = 4$

$c_R = 4$	Input shape ($m \times n \times b$)	Encoder mean time [s]	Encoder std time [s]	Decoder mean time [s]	Decoder std time [s]
A1D-CAE	1000, 1000, 63	6.34	0.061	4.45	0.027
NLPCA	1000, 1000, 63	2.73	0.085	2.65	0.033
SSCNet	1000, 1000, 63	0.12	0.021	0.07	0.005
SpatialNet	1000, 1000, 63	0.09	0.025	0.03	0.008
3D-CAE	1000, 1000, 63	0.35	0.051	0.30	0.009
PCA	1000, 1000, 63	0.40	0.037	0.58	0.028
k-means quantization	1000, 1000, 16	4.16	0.179	-	-
Huffman	16000000	7.14	0.397	21.65	2.134

time for the models with a compression factor of 4 are shown in Table XII. It can be seen that the purely spectral 1-D compression methods, such as A1D-CAE, NLPCA, and PCA, are slower compared to the 2-D compression methods, SSCNet and SpatialNet. The 2-D methods are faster because the architecture is not as deep as, for example, the A1D-CAE and because multiple spectral signatures are processed simultaneously. The 3D-CAE model takes longer to encode and decode than 2-D methods due to the additional spectral dimension but is faster than 1-D methods because the tiles process multiple spectral signatures simultaneously. Since most hyperspectral sensors are line scanners, the 1-D compression methods can be advantageous because compression can begin immediately after a line has been recorded. The 2-D and 3-D compression process can only start until several lines or the entire hyperspectral data cube has been recorded. The computational time did not consider the tiling and inverse tiling required for the 2-D compression process as preprocessing and postprocessing steps.

The computational time is also determined for the second compression stage consisting of k-means quantization and Huffman. Since the procedures are not optimized for GPUs, the calculation is carried out on an Intel[1] Xeon[1] Gold 6136 CPU @ 3.00 GHz.

V. CONCLUSION AND OUTLOOK

In this article, we propose CompNext1D as an advanced multistage lossy hyperspectral compression model that is transferable across multiple sensors. The adaptive extension of the A1D-CAE model allows the compression of spectral signatures from different hyperspectral sensors with deviating numbers of bands and characteristics. This transferability eliminates the requirement for individual models for each sensor. Furthermore, the A1D-CAE model can process spatially and temporally different datasets from the training data and the recording sensor. Currently, no state-of-the-art learning-based compression method is transferable to multiple other hyperspectral sensors. With the transferability of the A1D-CAE, we increase the utility of the increasing availability of large unlabeled hyperspectral data from space missions. As a result, the A1D-CAE can be trained as a backbone on a large amount of data with high spectral variability. The spectral-based input provides the model with more training data than the 2-D and 3-D models. This increased availability of training data ensures a more stable training process, resulting in a model with broad spectral variability and higher reconstruction accuracy. The pretrained model can be used as a solid backbone model for feature extraction and compressed domain analysis for onboard processing.

The evaluation shows that, for the different test datasets, DESIS, Greding, KHH, and PaviaU, a high reconstruction accuracy is achieved for the compression rate $c_R = 4$. Furthermore, all characteristic structures and colors are preserved, and the compression has no negative impact on the spatial dimension. The reconstruction accuracy, especially for higher compression rates, can be further improved by fine-tuning the pretrained A1D-CAE using a small amount of the limited target data.

The deeper architecture increases the ability to learn features and improves the generalization and transferability to unknown spatially independent and time-varying hyperspectral PRISMA data. This establishes a solid foundational understanding of hyperspectral data to learn more representative features, enhancing its robustness and generalization capabilities. Our presented basic model A1D-CAE and the extended model CompNext1D outperform the reference methods for all metrics and on all six test datasets for the compression rates $c_R = 4$ to $c_R \approx 100$. Furthermore, spectral signatures not present in the training data, such as ice and snow, can still be reconstructed with high reconstruction accuracy. The metrics show that A1D-CAE and CompNext1D achieve high reconstruction accuracy. The general spectral shape is reconstructed with high reconstruction accuracy for all compression rates. The visualization of the individual spectral signatures confirms the results of the evaluation metrics, such as SA, SNR, and PSNR. Since CompNext1D incorporates the A1D-CAE model in its architecture, it also has the property of sensor transferability with an additional increased compression rate due to the downstream compression stage. Furthermore, the model has the ability to compress hyperspectral data during capturing. This allows simultaneous data transmission and immediate onboard compressed analysis faster and more frequent data access.

In our future work, we will pretrain the A1D-CAE with datasets from space missions, fine-tune the model with a small amount of data from a different hyperspectral sensor, and evaluate the reconstruction accuracy compared to the base model and original data. Our goal is to create a flexible

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backbone based on an existing model that can be quickly fine-tuned to different sensors or target areas. The model from [55] features an intriguing design with skip connections facilitating information transfer across layers. Adding skip connections could enhance reconstruction accuracy, but the resulting tradeoff in compression rate must be considered. In addition, we will incorporate the spatial dimension into the model to further increase the compression rate while preserving structural information and color. Also, graph convolutional networks [56] and their interesting capabilities for capturing spatial–spectral feature representation could further improve the reconstruction accuracy. Using transformer architectures together with the large training database from the hyperspectral satellite missions could be promising to improve data compression [57].

APPENDIX
OVERVIEW OF THE DATASETS USED

The exact names of the training and test datasets are listed here to enhance reproducibility. The PRISMA datasets are available at [58]. Below is a list of the training and test datasets used.

PRISMA Training Datasets:

1) Italy (a) from the 2020.08.18 10:21:37: `PRS_L2B_STD_20200818102137_20200818102142_0001`.
2) Nigeria (b) from the 2020.11.12 10:12:31: `PRS_L2B_STD_20201112101231_20201112101235_0001`.
3) Sambia (c) from the 2020.07.31 08:49:42: `PRS_L2B_STD_20200731084942_20200731084946_0001`.
4) Germany (d) from the 2020.08.01 10:25:51: `PRS_L2B_STD_20200801102551_20200801102555_0001`.
5) India (e) from the 2021.05.15 04:47:52: `PRS_L2B_STD_20210515044752_20210515044756_0001`.
6) Portugal (f) from the 2020.08.05 11:35:27: `PRS_L2B_STD_20200805113527_20200805113531_0001`.

PRISMA Test Datasets:

1) Brazil (a) from the 2020.08.06, 13:41:07: `PRS_L2B_STD_20200806134107_20200806134111_0001`.
2) France (b) from the 2020.08.02, 10:43:55: `PRS_L2B_STD_20200802104355_20200802104359_0001`.
3) Italy (c) from the 2020.08.01, 10:27:17: `PRS_L2B_STD_20200801102717_20200801102721_0001`.
4) USA (d) from the 2021.03.01, 17:46:10: `PRS_L2B_STD_20210301174610_20210301174614_0001`.
5) Switzerland (e) from the 2020.08.07, 10:29:49: `PRS_L2B_STD_20200807102949_20200807102954_0001`.
6) Myanmar (f) from the 2021.05.08, 04:28:38: `PRS_L2B_STD_20210508042838_20210508042842_0001`.

The DESIS mission test dataset is made available in [59]. The exact name of the dataset recorded over Germany is `DESIS-HSI-L2A-DT0775159776_015-20221005T134326-V0215-SPECTRAL_IMAGE`.

The complete Greding data, including radiance and reflectance images, as well as ground-truth data, have been approved for sharing with the scientific community. The complete KHH data, including radiance and reflectance images, have been approved for sharing with the scientific community. Please contact the authors to get a copy.

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